

MULTIPLE CHOICE QUESTIONS

IN

HISTOLOGY & CELL BIOLOGY

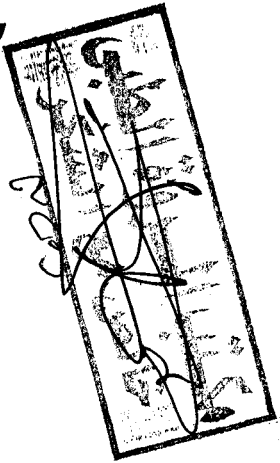
For

Second Year
Medical Students

By
Staff Members

Histology Department

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Nervous Tissue

(A) Multiple Choice Questions (MCQs)

Choose the best answer:

1. **Nissl bodies consist of:**
 - a. Clusters of synaptic vesicles.
 - b. Rough endoplasmic reticulum and ribosomes.
 - c. Golgi bodies.
 - d. Lysosomes and lipofuscin granules.
2. **Information is carried away from the neuron cell body by the:**
 - a. Axon.
 - b. Dendrite.
 - c. Myelin.
 - d. Ganglion.
3. **A special stain for myelin sheath is:**
 - a. PAS.
 - b. Osmic acid.
 - c. H&E.
 - d. Orcein.
4. **The common term "nerve fibers" refers to:**
 - a. Actin and myosin.
 - b. Axons.
 - c. Neurofilaments.
 - d. Microtubules.
5. **In neurons, which of the followings is related to the functional state?**
 - a. Neurofilaments.
 - b. Microtubules.
 - c. Nissl's granules.
 - d. Glycogen.
6. **Which of the following substances characteristically increases in abundance with increasing age in neurons:**
 - a. Neurofilaments.
 - b. Desmin.
 - c. Lipofuscin.
 - d. Urate oxidase.
7. **Which of the following is true for Nissl bodies?**
 - a. Are aggregates of residual bodies.
 - b. Undergo chromatolysis after injury to the axon.
 - c. Are found in axon hillock.
 - d. Are PAS positive.
8. **Which of the following is true for dendrites?**
 - a. Each has a constant diameter along its entire length.
 - b. Each ends with a typical terminal button.
 - c. They are covered by spines.
 - d. Are typically myelinated.
9. **Regarding muscle spindle:**
 - a. Efferent endings surrounding non striated portion of intrafusal muscle fibers.
 - b. Afferent endings surrounding the extrafusal muscle fibers.
 - c. Efferent endings surrounding striated portion of intrafusal muscle fibers.
 - d. Afferent endings surrounding the intrafusal muscle fibers and also the extrafusal muscle fibres.

- 10. Regarding myelin sheath, identify the correct statement:**
a. It is formed by oligodendrocytes in peripheral nervous system.
b. It is formed by astrocytes cells in central nervous system.
c. It is glycoprotein in nature.
d. It speeds up nerve impulse.
- 11. Bipolar neurons are present in:**
a. Vestibular ganglia.
b. Spinal ganglia.
c. Sympathetic ganglia.
d. Mesencephalic nucleus.
- 12. Small cells closely associated with neurons in peripheral ganglia are called:**
a. Satellite cells.
b. Pyramidal cells.
c. Postganglionic cells.
d. Preganglionic cells.
- 13. Which cell type forms the myelin sheath around myelinated axons in the central nervous system?**
a. Ependymal cell.
b. Schwann cell.
c. Oligodendrocyte.
d. Astrocyte.
- 14. Structures extending along the length of the axon which provide the substrate for axoplasmic transport are the:**
a. Nissl bodies.
b. Microtubules.
c. Synaptic vesicles.
d. Nodes of Ranvier.
- 15. What is the cell body of a neuron called?**
a. Ganglion.
b. Perikaryon.
c. Nissl bodies.
d. Terminal bouton.
- 16. What provides tensile strength to a neuron?**
a. Neurofilaments.
b. Myelinated nerve fibers.
c. Cell bodies.
d. Nodes of Ranvier.
- 17. Neuroglia cells which have phagocytic function are:**
a. Astrocytes.
b. Mesoglia.
c. Schwann cells.
d. Oligodendrocytes.
- 18. One of the following is classified as unencapsulated:**
a. Golgi tendon organ.
b. Meissner's corpuscles.
c. Krause end bulb.
d. Free nerve endings.
- 19. White matter appears white because of:**
a. The absence of blood vessels.
b. The absence of glial cells.
c. The presence of many collagen fibers.
d. The presence of many myelinated axons.

20. Which structure surrounds peripheral nerve fibres?
 - a. Trunk of primary dendrite.
 - b. Nissl bodies.
 - c. Perineurium.
 - d. Endoneurium.
21. Silver stain demonstrates:
 - a. Myelin sheath.
 - b. Nissle granules.
 - c. Early degenerated tracts.
 - d. Golgi bodies.
22. Myelinated nerve fibers without sheath of Schwann cells are found in:
 - a. Gray matter only.
 - b. White matter only.
 - c. Gray and white matter.
 - d. Peripheral nerve fibers.
23. The first change in retrograde degeneration is:
 - a. Fragmentation of neurofibrils.
 - b. Migration of the nucleus.
 - c. Chromatolysis.
 - d. Increase in lysosomes.
24. Golgi tendon organs are considered as:
 - a. Chemoreceptors.
 - b. Mechanoreceptors.
 - c. Photoreceptors.
 - d. Proprioceptors.
25. One of the following is considered as proprioceptor:
 - a. Krause's end bulb.
 - b. Pacinian corpuscle.
 - c. Meissner's corpuscles.
 - d. Ruffini's end organ.
26. Which type of neuroglia share in the formation of blood brain barrier?
 - a. Microglia.
 - b. Ependymal cells.
 - c. Astrocytes.
 - d. Oligodendrocytes.
27. Which type of neuroglia is responsible for movement of CSF?
 - a. Microglia.
 - b. Ependymal cells.
 - c. Astrocytes.
 - d. Oligodendrocytes.
28. One of the following is a connective tissue receptor:
 - a. Peritrichial nerve ending.
 - b. Muscle spindle.
 - c. Organ of Corti.
 - d. Krause's end bulb.

(B) Problem solving

1. A 9-year-old child was diagnosed as having an occipital brain tumor. A biopsy of this tumor mass reveals:
 - a. Marked hyperplasia of neurons.
 - b. Many degenerated neurons.
 - c. Multiple demyelinated nerve fibres.
 - d. Proliferation of macroglia.

2. Examination of immunohistochemically stained sections from the biopsy of the above case would reveal the following intermediate filament
 - a. Neurofilaments.
 - b. Cytokeratins.
 - c. Vimentin.
 - d. Glial fibrillar acidic protein.

3. A 30-year old man was diagnosed of a demyelinating disease. Which of the following stains are best used to demonstrate the changes in myelin:
 - a. Silver stain.
 - b. Methylene blue stain.
 - c. Osmic acid.
 - d. Potassium dichromate.

4. A 40-year-old house-wife was brought to the emergency room with injured ulnar nerve while using the kitchen knife. After surgical intervention, she was terribly concerned about the repair of this nerve. Proper regeneration of this injured nerve depends upon:
 - a. Approximation of two parts of axons.
 - b. Approximation of two parts of myelin sheath.
 - c. Approximation of two parts of the axon & the neurolemmal sheath.
 - d. Number of lymphocytes at site of injury.

(C) Match Column (A) with Column (B)

Table I

Column (A)	Column (B)
1. Spinal ganglion	a. Demonstrated by silver stain
2. Post synaptic membrane	b. Are rER and polyribosomes
3. Golgi apparatus	c. Cells are multipolar pyramidal
4. The axons	d. Has receptors for acetyl choline
5. Nissle bodies	e. Stained with PAS
	f. Have uniform diameter
	g. Cells are unipolar

Table II

Column (A)	Column (B)
1. The ulnar nerve	a. Osmic acid
2. The axon	b. Gap junctions
3. Site of contact of axon and cell body	c. Tapering
4. Myelin sheath is stained with	d. Silver
5. Electrical synapse	e. Has myelinated nerve fibers
	f. Axosomatic
	g. Is devoid of Nissl granules

Table III

Column (A)	Column (B)
1. In regeneration	a. Substantia nigra
2. Purkinji cells are found in	b. Shows delicate fibers or granules
3. Melanin pigment is seen in	c. Cerebellum
4. The synaptic cleft	d. The grey mater
5. Pyramidal cells are seen in	e. Cerebral cortex
	f. There is proliferation of Schwann cells
	g. Are sensory neurons

Table IV

Column (A)	Column (B)
1. In spinal ganglion	a. Nerve fibers are thin non-myelinated
2. Centerioles in neurons	b. Silver
3. Myelin sheath is stained by	c. Sudan III
4. Golgi is stained by	d. Cells are bipolar
5. In sympathetic ganglia	e. Are absent
	f. Osmic acid
	g. Satellite cells are abundant

Table V

Column (A)	Column (B)
1. Astrocytes	a. Mesodermal in origin
2. Oligodendrocytes	b. Simple cubical ciliated.
3. Microglia	c. Star shaped with multiple processes.
4. Schwann cells	d. Around nerve cells in ganglia
5. Ependymal cells	e. Form myelin in central nervous system
	f. In peripheral nervous system
	g. Contain Nissl's granules

Answers:

A. MCQ

1	b	8	c	15	b	22	b
2	a	9	c	16	a	23	c
3	b	10	d	17	b	24	d
4	b	11	a	18	d	25	b
5	c	12	a	19	d	26	c
6	c	13	c	20	d	27	b
7	b	14	b	21	d	28	d

B. Problem solving

1	2	3	4
d	d	c	c

C. Matching

Table I

1	2	3	4	5
g	d	a	f	b

Table II

1	2	3	4	5
e	g	f	a	b

Table III

1	2	3	4	5
f	c	a	b	e

Table IV

1	2	3	4	5
g	e	f	b	a

Table V

1	2	3	4	5
c	e	a	f	b

Central Nervous System

(A) Multiple Choice Questions (MCQs)

Choose the best answer:

1. **The central nervous system is soft because:**
 - a. It consists of neurons joined by connective tissue (C.T.).
 - b. It is rich in blood vessels.
 - c. It does not contain C.T.
 - d. It contains fat cells.
2. **Which of the following is true for central nervous system?**
 - a. Rich in C.T. cells.
 - b. Soft jelly like.
 - c. Has grey matter & white matter.
 - d. No nerve fibers
3. **Distribution of grey and white matter depends on the distribution of:**
 - a. Blood vessels.
 - b. Myelin.
 - c. Pigments.
 - d. Astrocytes.
4. **The central canal is lined by epithelium:**
 - a. Simple squamous.
 - b. Stratified squamous.
 - c. Simple cuboidal partially ciliated.
 - d. Simple columnar non ciliated.
5. **One of the followings is not present in grey matter:**
 - a. Nerve cells.
 - b. Myelinated nerve fibres.
 - c. Neuroglial cells.
 - d. Unmyelinated nerve fibres.
6. **Which of the following is a character of thoracic level of spinal cord?**
 - a. Oval in shape.
 - b. Large in size.
 - c. It has 4 horns.
 - d. It has 2 lateral horns.
7. **One of the following tracts carries sensations to subcortical level:**
 - a. Dorsal spinocerebellar tract.
 - b. Gracile tract.
 - c. Lateral spinothalamic.
 - d. Ventral spinothalamic.
8. **Receptor/s for proprioception is/are:**
 - a. Pacinian corpuscle.
 - b. Meissner's corpuscle.
 - c. Ruffini' corpuscle.
 - d. Meissner's and Ruffini' corpuscles.
9. **Which of the following is a receptor for vibration?**
 - a. Meissner's corpuscle.
 - b. Ruffini's corpuscle.
 - c. Merkel's disc.
 - d. Pacinian corpuscle.
10. **The first order neuron in proprioception from lower half of body is in:**
 - a. Gracile nucleus.
 - b. Cuneate nucleus.
 - c. Spinal ganglia.
 - d. Main sensory nucleus.

11. **The first order neuron in proprioception from upper half of body is in:**
 - a. Gracile nucleus.
 - b. Cuneate nucleus.
 - c. Spinal ganglia.
 - d. Main sensory nucleus.
12. **External arcuate fibres carry impulses to:**
 - a. Thalamus.
 - b. Cerebral cortex.
 - c. Cerebellum.
 - d. Pons.
13. **Internal arcuate fibres are present in:**
 - a. Superior midbrain.
 - b. Inferior midbrain.
 - c. Open medulla.
 - d. Closed medulla.
14. **The second order neuron in proprioception from lower half of body is in:**
 - a. Gracile nucleus on same side.
 - b. Cuneate nucleus on same side.
 - c. Gracile nucleus on opposite side.
 - d. Cuneate nucleus on opposite side.
15. **One of the following is true about gracile tract:**
 - a. Presents only in cervical & thoracic levels of spinal cord.
 - b. Carries proprioception from the upper half of the body.
 - c. Terminates at gracile nucleus in medulla.
 - d. Arises from spinal ganglia in upper half of body.
16. **One of the following is true about cuneate tract:**
 - a. It arises from spinal ganglia in upper half of body.
 - b. It arises from cuneate nucleus on same side.
 - c. It arises from cuneate nucleus on opposite side.
 - d. It is present in lateral column of white matter in spinal cord.
17. **Arcuate fibres carry:**
 - a. Proprioception from the body.
 - b. Pain and temperature from head.
 - c. Pain and temperature from the body.
 - d. Crude touch from head.
18. **Internal arcuate fibres:**
 - a. Arise from gracile and cuneate nuclei.
 - b. Terminate at gracile and cuneate nuclei.
 - c. Form lateral lemniscus.
 - d. Carry proprioceptive impulses from head.
19. **Origin of medial lemniscus is:**
 - a. Gracile and cuneate nuclei of same side.
 - b. Gracile and cuneate nuclei of opposite side.
 - c. Arcuate nucleus of same side.
 - d. Arcuate nucleus of opposite side.

- 20. Decussation of internal arcuate fibres takes place in:**
- Spinal cord.
 - Upper part of closed medulla.
 - Lower part of closed medulla.
 - Open medulla.
- 21. The following is true about first order neurons in proprioception of body:**
- They are the small cells in spinal ganglia.
 - They have thin unmyelinated nerve fibres.
 - Central branches enter spinal cord through lateral division of posterior roots.
 - They give ascending and descending branches in spinal cord.
- 22. Which of the following is involved in the blood brain barrier?**
- Astrocytes.
 - Ependymal cells.
 - Oligodendrocytes.
 - Schwann cells.
- 23. First order neurons in proprioception pathway from face and head are in:**
- Trigeminal ganglion.
 - Dorsal root ganglion.
 - Spinal N. of trigeminal n.
 - Mesencephalic N. of trigeminal n.
- 24. First order neurons for pain and temperature from face are located in:**
- Trigeminal ganglion.
 - Dorsal root ganglion.
 - Mesencephalic nucleus of trigeminal nerve.
 - Spinal nucleus of trigeminal nerve.
- 25. Second order neuron in simple touch pathway from face and head are in:**
- Trigeminal ganglion.
 - Main sensory nucleus.
 - Spinal nucleus of trigeminal nerve.
 - Mesencephalic nucleus of trigeminal nerve.
- 26. Trigeminal lemniscus is present in:**
- Medulla and Pons.
 - Pons and mid brain.
 - Mid brain.
 - Medulla.
- 27. The arachnoid barrier is present between:**
- Extracerebral capillaries & subarachnoid space.
 - Astrocytes.
 - Blood and brain cells.
 - Subarachnoid space and CSF.
- 28. One of the following is a component of Blood CSF barrier:**
- Border cells of arachnoid matter.
 - tight junctions between ependymal cells.
 - astrocytes.
 - Trabecular cells of arachnoid matter.

29. **Blood brain barrier is formed of tight junctions between:**
 a. fenestrated endothelium. b. End feet of astrocytes.
 c. non fenestrated endothelium. d. Border cells.
30. **Border cells can be found in:**
 a. pia matter. b. arachnoid matter. c. dura matter. d. pons.
31. **Pia matter is formed of:**
 a. simple squamous epithelium. b. simple columnar epithelium.
 c. simple columnar ciliated epithelium. d. modified fibroblasts.
32. **One of the following is true about climbing fibres:**
 a. They arise from pontine nuclei.
 b. They synapse with granular cells.
 c. Each fibre synapse with one Purkinje cell.
 d. They carry inhibitory signals.

(B) Problem solving

1. **A 5-year-old girl with meningitis. Doctor prescribes her antibiotics. Which of the following statements should be considered true?**
 a. Blood brain barrier (B.B.B.) normally allows the passage of all antibiotics.
 b. The B.B.B. does not allow passage of antibiotics due to big molecular size.
 c. Systemic administration of penicillin normally → large amount entering CNS.
 d. Meninges become more permeable to penicillin at the site of inflammation and its concentration rises in cerebrospinal fluid.
2. **A patient with lesion in right gracile nucleus is expected to have less.....**
 a. tactile localization at right arm.
 b. Fine touch at left leg.
 c. stereognosis at right hand.
 d. tactile discrimination at right leg.
3. **An injury to the posterior column of the lumbar spinal cord may be manifested as loss of ...**
 a. Fine touch at upper limbs.
 b. tactile discrimination at lower limbs.
 c. Crude touch at upper limbs.
 d. Pain & temperature at the lower limbs.

- (C) Match column A with column B**

Answers:

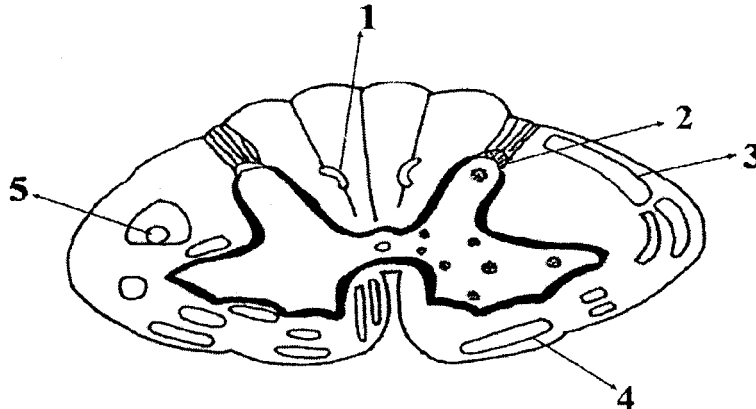
B. Problem solving

C. Matching

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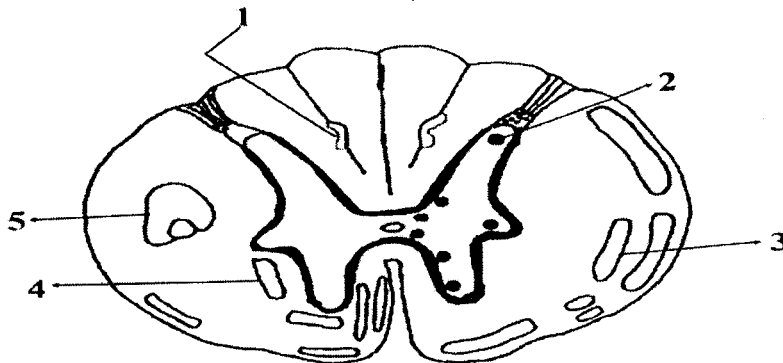
CNS LEVELS

1) Match column A with column B:



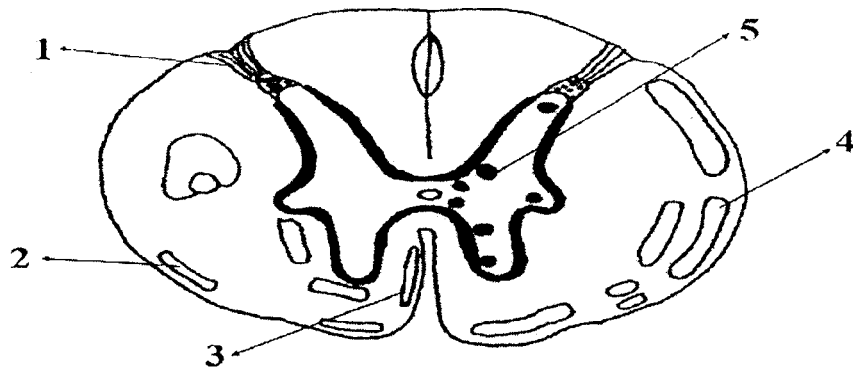
Column A	Column B
1) Label 1...	a) Originates from Clark's nucleus
2) Label 2...	b) Second order neuron for pain & temperature.
3) Label 3...	c) Completes the reflex arc.
4) Label 4...	d) Forms the ventral tegmental decussation.
5) Label 5...	e) Originates from main sensory nucleus.
	f) Forms the dorsal tegmental decussation.
	g) Second order neuron for fine touch.

2) Match column A with column B:



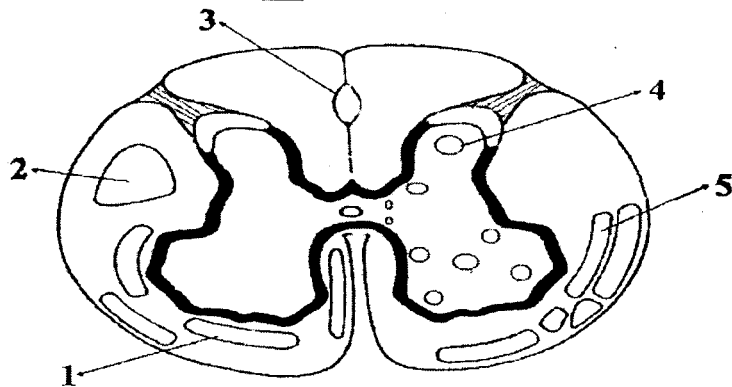
Column A	Column B
1) Label 1...	a) Arises from red nucleus.
2) Label 2...	b) Controls the voluntary movements.
3) Label 3...	c) Continues as spinal lemniscus.
4) Label 4...	d) Gives rise to ventral spinothalamic tract.
5) Label 5...	e) Gives rise to lateral spinothalamic tract.
	f) Descending fibers from cuneate tract.
	g) Arises from reticular formation.

3) Match column A with column B:



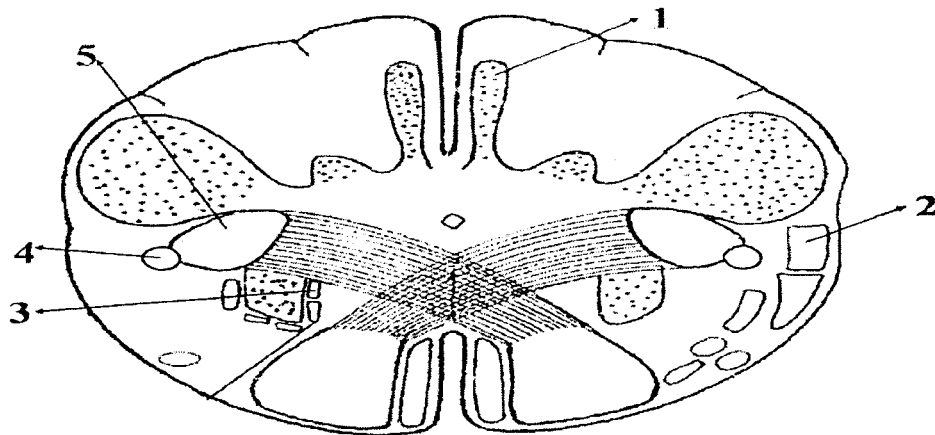
Column A	Column B
1) Label 1...	a) Second order neuron for unconscious proprioception from same & opposite sides of the lower limb.
2) Label 2...	b) First order neuron for pain & temperature.
3) Label 3...	c) Second order neuron for unconscious proprioception from same side of the body.
4) Label 4...	d) Arises from lateral vestibular nucleus.
5) Label 5...	e) Arises from medial vestibular nucleus.
	f) Second order neuron for pain & temperature.
	g) Continuation of medial longitudinal bundle.

4) column A with column B:



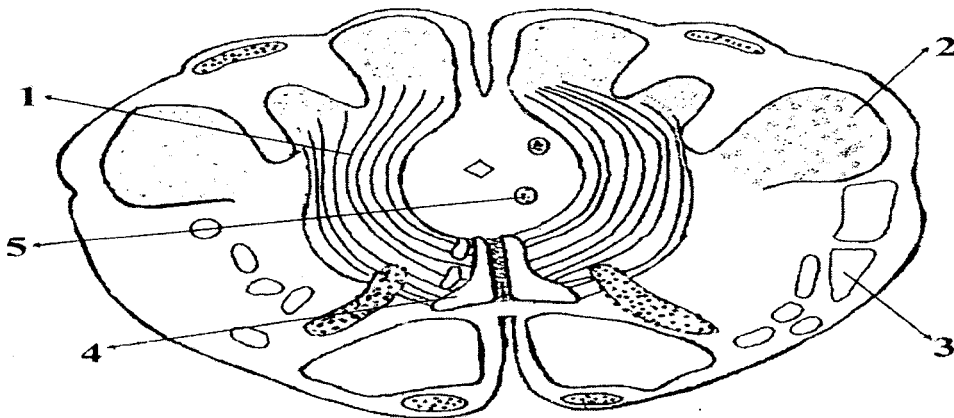
Column A	Column B
1) Label 1...	a) Completes the reflex arc.
2) Label 2...	b) Continuation of the medial longitudinal bundle.
3) Label 3...	c) Second order neuron for crude touch.
4) Label 4...	d) Part of the corticobulbar tract.
5) Label 5...	e) Part of the corticospinal tract.
	f) Arises from the reticular formation.
	g) Second order neuron for pain & temperature.

5) Match column A with column B:



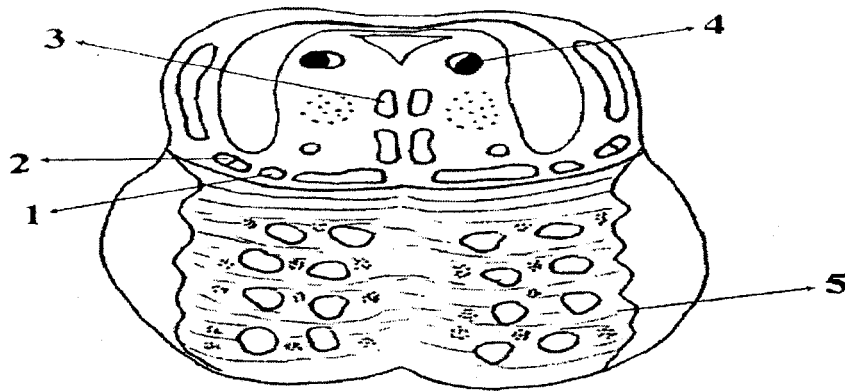
Column A	Column B
1) Label 1...	a) Continues as septomarginal tract.
2) Label 2...	b) Originates from upper two-thirds of area 4.
3) Label 3...	c) Originates from lower one-third of area 4.
4) Label 4...	d) Forms the ventral tegmental decussation.
5) Label 5...	e) Continues as sulcomarginal tract.
	f) 2 nd order neuron for proprioception to subcortical level.
	g) Second order neuron for proprioception to cerebral cortex.

6) Match column A with column B:



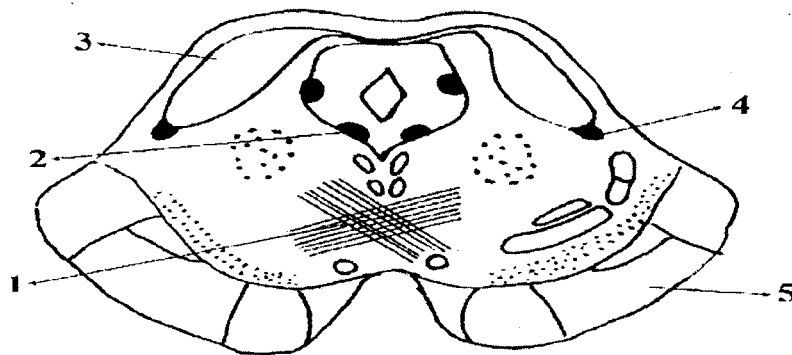
Column A	Column B
1) Label 1...	a) 2 nd order neuron for unconscious proprioception.
2) Label 2...	b) Nucleus of hypoglossal nerve.
3) Label 3...	c) Nucleus of vagus nerve.
4) Label 4...	d) Second order neuron for fine touch.
5) Label 5...	e) Forms the sensory decussation.
	f) Axons join trigeminal lemniscus.
	g) Forms the motor decussation.

7) Match column A with column B:



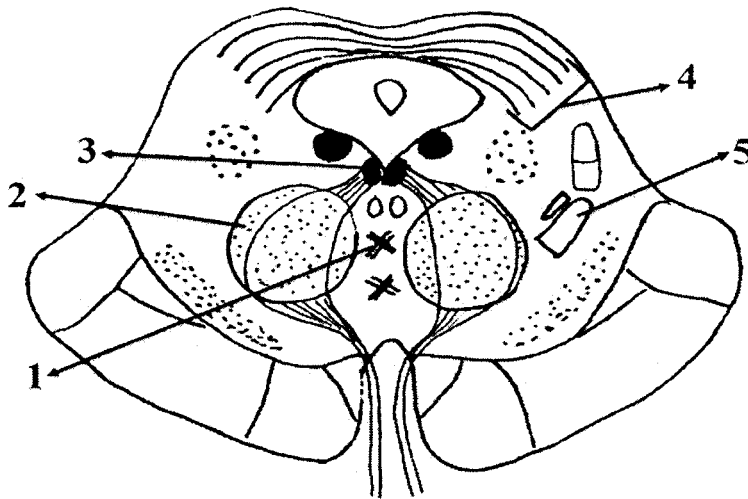
Column A	Column B
1) Label 1...	a) Separates the pyramidal tract bundles.
2) Label 2...	b) First order neuron for crude touch.
3) Label 3...	c) Ends in VPMN of the thalamus.
4) Label 4...	d) Continues as septomarginal tract.
5) Label 5...	e) Continues as sulcomarginal tract.
	f) First order neuron for fine touch from the face.
	g) Ends in VPLN of the thalamus.

8) Match column A with column B:



Column A	Column B
1) Label 1...	a) Medial lemniscus.
2) Label 2...	b) Center for auditory reflexes.
3) Label 3...	c) Originates from upper two-thirds of area 4.
4) Label 4...	d) Lateral lemniscus.
5) Label 5...	e) Nucleus of oculomotor nerve.
	f) Decussation of superior Cerebellar peduncles.
	g) Nucleus of trochlear nerve.

9) Match column A with column B:



Column A	Column B
1) Label 1...	a) Trigeminal lemniscus.
2) Label 2...	b) Center for visual reflexes.
3) Label 3...	c) Medial lemniscus.
4) Label 4...	d) Decussation of rubrospinal tracts.
5) Label 5...	e) Decussation of tectospinal tracts.
	f) Red nucleus.
	g) Oculomotor nucleus.

Model Answer

1)

1	2	3	4	5
c	b	a	e	d

2)

1	2	3	4	5
f	d	c	g	b

3)

1	2	3	4	5
b	d	g	a	c

4)

1	2	3	4	5
f	e	a	c	g

5)

1	2	3	4	5
g	f	e	d	b

6)

1	2	3	4	5
e	f	a	d	b

7)

1	2	3	4	5
c	g	e	f	a

8)

1	2	3	4	5
f	g	b	d	c

9)

1	2	3	4	5
e	f	g	b	c

Eye

(A) Multiple Choice Questions (MCQs)

Choose the best answer:

- 1. Concerning the anterior limiting membrane:**
 - a. It is a thin membrane.
 - b. It is a cellular layer.
 - c. It can regenerate.
 - d. It forms a protective barrier to the cornea.
- 2. The corneal epithelium has the following characteristic:**
 - a. Keratinized stratified squamous epithelium.
 - b. Contains many myelinated nerve endings.
 - c. Shows few mitotic figures.
 - d. The superficial layer has microvilli.
- 3. Following trauma, a person will develop corneal opacity caused by:**
 - a. Injury of Descemet's membrane.
 - b. Injury of Bowman's membrane.
 - c. Injury of posterior limiting membrane.
 - d. Injury of corneal epithelium.
- 4. Which statement regarding the sclera is correct?**
 - a. Forms the posterior five sixths of middle coat.
 - b. Has many keratocytes.
 - c. Ciliary muscle fibers are inserted in.
 - d. It is relatively avascular.
- 5. At the corneo-scleral junction:**
 - a. Bowman's membrane forms the spaces of Fontana.
 - b. Descemet's membrane becomes continuous in the sclera.
 - c. Conjunctival epithelium is keratinized squamous epithelium.
 - d. Stem cells of corneal epithelium are concentrated.
- 6. A patient diagnosed as a case of glaucoma suffers from:**
 - a. Corneal opacity.
 - b. Abnormal corneal curvature.
 - c. Increased intraocular pressure.
 - d. Loss of lens transparency.
- 7. Which of the following shares in the formation of the tear film:**
 - a. Goblet cells of the conjunctiva.
 - b. Moll's glands.
 - c. Choroid.
 - d. Ciliary body.

- 8. The lacrimal sac is lined by:**
- Stratified columnar epithelium.
 - Pseudo stratified columnar epithelium.
 - Simple columnar epithelium.
 - Pseudo stratified columnar ciliated epithelium.
- 9. The lacrimal gland is classified as being:**
- Simple alveolar serous gland.
 - Compound tubulo-alveolar mucous gland.
 - Compound tubulo-alveolar serous gland.
 - Simple branched tubulo-alveolar serous gland.
- 10. The tarsal plate of the eye lid contains:**
- Meibomian glands.
 - Glands of Zeis.
 - Glands of Moll.
 - Lacrimal glands.
- 11. The subcapsular epithelium of the eye lens is:**
- Simple squamous covers the anterior surface only.
 - Simple cuboidal covers the anterior surface only.
 - Simple cuboidal covers the whole lens.
 - Simple cuboidal covers the posterior surface only.
- 12. Bruch's membrane is formed of:**
- Basement membranes of choriocapillary lamina, pigmented epithelium, two collagen layers with elastic fibers in between.
 - Basement membranes of suprachoroidal lamina and pigmented epithelium, two collagen layers with elastic fibers in between.
 - Two collagen layers with elastic fibers in between.
 - Two layers of elastic fibers with a collagen layer in between.
- 13. Regarding the retinal pigmented epithelium:**
- Single layer of high columnar cells.
 - Allows reflection of light.
 - Outer part is rich in melanin pigments.
 - Shares in formation of blood retinal barrier.
- 14. The outer nuclear layer of retina contains nuclei of:**
- Rods and cones.
 - Müller cells.
 - Bipolar nerve cells.
 - Ganglion cells.
- 15. The inner nuclear layer of retina contains nuclei of:**
- Rods & cones, amacrine cells, Müller cells and bipolar cells.
 - Rods & cones, amacrine cells, Müller cells and horizontal cells.
 - Amacrine cells, Müller cells, ganglion cells and bipolar cells.
 - Amacrine cells, Müller cells, horizontal cells and bipolar cells.

16. The outer limiting membrane is formed by junctional complexes:

- a. Between photoreceptors and bipolar nerve cells.
- b. Between photoreceptors and Müller cells.
- c. Between Müller cells and amacrine cells.
- d. Between Müller cells.

17. The inner limiting membrane is formed by junctions between:

- a. End processes of Müller cells.
- b. Müller cells and bipolar cells.
- c. Müller cells and ganglion cells.
- d. Amacrine cells.

18. The outer plexiform layer is formed by synapses between:

- a. Rods, cones, bipolar nerve cells and horizontal cells.
- b. Rods, cones, Müller cells and bipolar nerve cells.
- c. Rods, cones, ganglion cells and horizontal cells.
- d. Rods, cones, bipolar nerve cells and amacrine cells.

19. Concerning Muller cells:

- a. Share in formation of outer limiting membrane only.
- b. Share in formation of inner limiting membrane only.
- c. Their nuclei are present in the outer nuclear layer.
- d. Their nuclei are present in the inner nuclear layer.

20. The following cells are glial cells:

- a. Horizontal cells.
- b. Amacrine cells.
- c. Müller cells.
- d. Rods.

21. The fovea centralis:

- a. Is highly vascular.
- b. Contains rods only.
- c. Formed of less number of layers.
- d. Present at the periphery of the blind spot.

22. The characteristics of cones include:

- a. They are about 70 million.
- b. They are of one type.
- c. They exist in fovea.
- d. The discs are continuously phagocytosed.

23. Regarding the macula lutea:

- a. Elevated spot opposite center of lens.
- b. Depressed spot opposite center of lens.
- c. Fovea centralis is present at its periphery.
- d. White depressed spot.

- 24. A patient presenting by redness and watering of eye, known clinically as pink eye is caused by:**
- a. Inflammation of Zeis gland.
 - b. Inflammation of Meibomian gland.
 - c. Inflammation of Moll's gland.
 - d. Inflammation of conjunctiva.

(B) Problem solving

- 1. A patient presenting with corneal opacity causing visual defects was directed to corneal transplantation is having:**
 - a. Injured epithelium.
 - b. Corneal vascularization.
 - c. Irregular substantia arrangement.
 - d. Injured Bowman's membrane.
- 2. A patient presenting with acute eye pain, by examination showed rise of the intraocular pressure, a condition known as...due to...:**
 - a. Cataract, lens opacity.
 - b. Presbyopia, decreased lens elasticity.
 - c. Glaucoma, increased vitreous formation.
 - d. Glaucoma, improper drainage of aqueous.
- 3. A 50-year-old patient presented with difficulty to focus during reading, was diagnosed as...due to....:**
 - a. Cataract, lens opacity.
 - b. Cataract, denaturated crystallin.
 - c. Presbyopia, defective accommodation.
 - d. Presbyopia, defective lens fibers.
- 4. A 20-year-old female patient presenting with painless visual defects and normal intraocular pressure was diagnosed as...**
 - a. Cataract.
 - b. Presbyopia.
 - c. Glaucoma.
 - d. Retinal detachment.
- 5. A 60-year-old male patient presenting with painless central visual defects and normal intraocular pressure was diagnosed as...**
 - a. Cataract.
 - b. Presbyopia.
 - c. Glaucoma.
 - d. Macular degeneration.
- 6. A twenty-year-old female complaining of a chalazion in her eye lid, the cause may be due to:**
 - a. Inflammation of Zeis gland.
 - b. Inflammation of lacrimal gland.
 - c. Inflammation of Moll's gland.
 - d. Inflammation of Meibomian gland.

7. An old man complaining of a small swelling at the margin of the eyelid the most probable origin is:
- Meibomian gland.
 - Zeis gland.
 - Lacrimal gland.
 - Tarsal gland.
8. A patient presenting with corneal opacity due to injury of the anterior limiting membrane, the condition could be treated by:
- Eye ointments.
 - Cataract surgery.
 - Lasik surgery.
 - Corneal transplantation.
9. A young man presenting with defective vision, by examination retinal detachment was found the possible cause is:
- Separation of rods and cones.
 - Separation of retinal pigmented epithelium.
 - Separation of Bruch's membrane.
 - Separation of outer limiting membrane.

(C) Match column A with column B

Table I

Column (A)	Column (B)
1. Cornea	a. Simple columnar epithelium
2. Lacrimal duct	b. Non keratinized stratified squamous epithelium
3. Conjunctiva	c. Keratinized stratified squamous epithelium
4. Subcapsular epithelium of the lens	d. Simple cuboidal epithelium
5. Outer surface of eye lid	e. Pseudo stratified columnar ciliated epithelium
	f. Stratified columnar epithelium with goblet cells
	g. Simple cubical epithelium

Table II: At the corneo-scleral junction

Column (A)	Column (B)
1. Corneal epithelium	a. Forms the pectinate ligament
2. Bowman's membrane	b. Replaced by subconjunctival C.T
3. Collagen lamellae	c. Lines trabecular meshwork
4. Microvasculature	d. Reflected as bulbar conjunctiva
5. Descemet's endothelium	e. Metabolite distribution
	f. Secretes aqueous humor
	g. Become irregular with appearance of canal of Schlemm

Answers:

A. MCQ

1	d	6	c	11	b	16	b	21	c
2	d	7	a	12	a	17	a	22	c
3	b	8	d	13	d	18	a	23	b
4	d	9	c	14	a	19	d	24	d
5	d	10	a	15	d	20	c		

B. Problem solving

1	2	3	4	5	6	7	8	9
d	d	c	d	d	d	b	d	b

C. Matching

Table I

1	2	3	4	5
b	e	f	d	c

Table II

1	2	3	4	5
d	b	g	e	c

Ear

(A) Multiple Choice Questions (MCQs)

Choose the best answer:

- 1. The auricle contains a plate of:**
 - a. Elastic cartilage.
 - b. Hyaline cartilage.
 - c. Yellow elastic connective tissue.
 - d. White fibrocartilage.
- 2. Concerning the auditory meatus:**
 - a. Belongs to middle ear.
 - b. Is formed of bone.
 - c. Exhibits ceruminous glands.
 - d. Exhibits hairs in its inner part.
- 3. Ceruminous glands are:**
 - a. Coiled tubular apocrine sweat glands.
 - b. Coiled tubular holocrine sweat glands.
 - c. Coiled tubular apocrine sebaceous glands.
 - d. Coiled tubular holocrine sebaceous glands.
- 4. The outer surface of the tympanic membrane is lined with:**
 - a. Stratified squamous epithelium.
 - b. Simple columnar epithelium.
 - c. Simple cuboidal epithelium.
 - d. Pseudo stratified squamous epithelium.
- 5. The inner surface of the tympanic membrane is lined with:**
 - a. Stratified squamous epithelium.
 - b. Simple columnar epithelium.
 - c. Simple cuboidal epithelium.
 - d. Pseudo stratified columnar epithelium.
- 6. Concerning the Shrapnel's membrane, one is true:**
 - a. It is the inferior part of the tympanic membrane.
 - b. It is opaque.
 - c. It is firm.
 - d. It lacks collagen fibers.
- 7. The middle ear is characterized by:**
 - a. The lateral wall is closed by the 2ry tympanic membrane.
 - b. The oval window is closed by the 2ry tympanic membrane.
 - c. The round window represents the lateral wall.
 - d. Posteriorly, it is related to mastoid air cells.

8. **As regards the Eustachian tube:**
 - a. Its cartilaginous part is lined by simple cuboidal epithelium.
 - b. It is lined with pseudostratified columnar ciliated epithelium.
 - c. It is always open.
 - d. Connects the middle ear to the oro-pharynx.
9. **The bony labyrinth-----:**
 - a. is filled with perilymph.
 - b. is formed of cochlear duct and membranous semicircular canals.
 - c. encloses stria vascularis.
 - d. houses the utricle and saccule.
10. **The membranous labyrinth includes:**
 - a. The cochlea.
 - b. The vestibule.
 - c. The utricle.
 - d. The semicircular canals.
11. **Structural features of the cochlea include:**
 - a. It is a bony tube forming $2\frac{1}{2}$ turns.
 - b. It turns around a membranous axis.
 - c. The modulus contains cell bodies of vestibular neurons.
 - d. It is a part of membranous labyrinth.
12. **Concerning the organ of Corti:**
 - a. It is present in the scala tympani.
 - b. It rests on the vestibular membrane.
 - c. It is bathed in endolymph.
 - d. It is responsible for sensation of equilibrium.
13. **As regards the cochlear duct:**
 - a. It is the upper compartment of the cochlea.
 - b. It contains the macula.
 - c. Stria vascularis forms the lower border.
 - d. The roof is formed by the vestibular membrane.
14. **The endolymph of the cochlea is:**
 - a. Present in the scala vestibule.
 - b. Present in the scala tympani.
 - c. Secreted by stria vascularis.
 - d. Of high Na^+ and K^+ content.
15. **The organ of Corti is characterized by:**
 - a. It is supported by tectorial membrane.
 - b. It is a neuro receptor for balance.
 - c. It is present in the scala vestibuli.
 - d. Covered by the basilar membrane.
16. **Innermost supporting cells of organ of Corti are:**
 - a. Phalangeal cells.
 - b. Hensen's cells.
 - c. Cells of Claudius.
 - d. Border cells.

17. Concerning phalangeal cells:

- a. The inner cells have upper convex ends.
- b. They enclose the tunnel of Corti.
- c. They include 3 rows of inner cells.
- d. The outer cells have a cytoplasmic phalanx.

18. Concerning the pillar cells:

- a. The inner pillar cells form 1 row, the outer cells from 3 to 4 rows.
- b. Each outer pillar cell gives off a long process.
- c. They contain keratin bundles.
- d. They are in direct contact with hair cells.

19. Concerning the inner hair cells of the organ of Corti:

- a. Form three rows of cells.
- b. Supported by inner phalangeal cells.
- c. Covered by a membrane containing calcium carbonate crystals.
- d. Contain many true cilia.

20. The macula demonstrates:

- a. Hair cells with one true cilium only.
- b. Cubical supporting cells with apical microvilli.
- c. A covering otolithic membrane.
- d. A covering cupula.

21. A person presenting with sensory hearing loss is expected to suffer from:

- a. Inflammation of the meatus.
- b. Deformed ossicles.
- c. Perforation of tympanic membrane.
- d. Diseases of inner ear.

22. A patient having infection spread from nasopharynx to middle ear. The spread is expected to be via:

- a. Ossicles.
- b. Chorda tympani.
- c. Eustachian tube.
- d. Oval window.

(B) Problem solving:

1. A 10 years old child complained of loss of hearing following infection associated with external ear inflammation, by examination he showed:

- a. Defective hair cells of macula.
- b. Defective hair cells of crista.
- c. Conductive hearing loss.
- d. Defective supporting cells.

2. A 7 years old girl presented with earache, ear discharge and history of recent sore throat, by examination.....was found:

- a. Sensorineural hearing loss.
- b. Perforated tympanic membrane.
- c. cochlear neuritis.
- d. Defective malleus.

3. A 4 years male patient presented with permanent hearing loss following examination cochlear transplantation was decided, the likely cause was:
 - a. Supporting cells damage.
 - b. Macula hair cells damage.
 - c. Crista hair cells damage.
 - d. Organ hair cells damage.
4. A middle-aged woman came to the clinic suffering from vertigo, a possible cause explaining her status may be:
 - a. Cochlear nerve degeneration.
 - b. Loss of hair cells of organ of Corti.
 - c. Otitis media.
 - d. Defective vestibular apparatus.
5. Degeneration of the cochlear nerve in the inner ear causes:
 - a. Motion sickness.
 - b. Neural deafness.
 - c. Vertigo.
 - d. Conductive deafness.
6. A child three years old pushed an ear cotton bud forcibly along his external auditory canal. He was diagnosed as:
 - a. Motion sickness.
 - b. Vertigo.
 - c. Neural deafness.
 - d. Conductive deafness.
7. Neural deafness diagnosed in children can be caused by drugs which damage...
 - a. Hair cells in organ of Corti.
 - b. External auditory canal.
 - c. Auditory ossicles.
 - d. Chorda tympani nerve.

(C) Match column A with column B

Table I

Column (A)	Column (B)
1. Outer surface of tympanic membrane	a. Handle of malleus
2. Inner surface of tympanic membrane	b. No collagen fibers
3. Pars flaccida of tympanic membrane	c. Stratified squamous epithelium
4. Collagen fibers of tympanic membrane	d. Radial and circular
5. Attached to the tympanic membrane	e. Three layers of collagen
	f. Simple cuboidal epithelium
	g. Foot plate of the stapes

Table II:

Column (A)	Column (B)
1. Oval window	a. Tensor tympani
2. Round window	b. Stapedius
3. Malleus	c. Mastoid air cells
4. Anterior wall of tympanic cavity	d. Stapes footplate
5. Posterior wall of tympanic cavity	e. Secondary tympanic membrane
	f. Eustachian tube
	g. Chorda tympani

Table III:

Column (A)	Column (B)
1. Supporting cells	a. Cupula
2. Tunnel of Corti	b. Phalangeal cells
3. Perilymph	c. Pillar cells
4. Crista ampullaris	d. Tectorial membrane
5. Macula	e. Otolithic membrane
	f. Bony semicircular canals
	g. Spiral ganglia

Answers:**A. MCQ**

1	a	7	d	13	d	19	b
2	c	8	b	14	c	20	c
3	a	9	a	15	a	21	d
4	a	10	c	16	d	22	c
5	c	11	a	17	d		
6	d	12	c	18	c		

B. Problem solving

1	2	3	4	5	6	7
c	b	d	d	b	d	a

C. Matching

Table I

1	2	3	4	5
c	f	b	d	a

Table II

1	2	3	4	5
d	e	a	f	c

Table III

1	2	3	4	5
b	c	f	a	e

Digestive System

I) Digestive Tract

(A) Multiple Choice Questions (MCQs)

Choose the best answer:

1. Which organ is covered by partially keratinized Stratified Squamous epithelium?
 - a. The soft palate.
 - b. Floor of the mouth.
 - c. Inner aspect of the cheeks.
 - d. Dorsum of the tongue.
2. A section in the lip red margin contains:
 - a. Hairy thin skin.
 - b. Sweat glands.
 - c. Rich sensory innervations.
 - d. Scanty blood capillaries.
3. The lip sections contain:
 - a. Buccinators skeletal muscle.
 - b. Compound acinar labial gls.
 - c. A vermillion.
 - d. Thick skin.
4. The dorsum of the tongue has:
 - a. Fully keratinized epithelium.
 - b. Lingual tonsils on the ant. 2/3.
 - c. No tongue papillae.
 - d. Firmly adherent mucosa.
5. The filiform papillae:
 - a. Contain few taste buds.
 - b. Are red in color.
 - c. Are covered by keratinized stratified squamous epithelium.
 - d. Cover post. 2/3 of the tongue.
6. The filiform papillae:
 - a. The least common papillae.
 - b. Covered by non- keratinized stratified squamous epithelium.
 - c. Cover the anterior 2/3 of the tongue.
 - d. Have numerous taste buds.
7. The fungiform papillae are characterized by:
 - a. Covered by keratinized stratified squamous epithelium.
 - b. Red appearance.
 - c. Scattered on post. 2/3 of the tongue.
 - d. In front of sulcus terminalis only.

8. **The fungiform papillae:**
 - a. Have no taste buds.
 - b. Covered by simple squamous epithelium.
 - c. Richly vascularized C.T. core.
 - d. Only 6-14 papillae.
9. **Concerning the circumvallate papillae one of the following is true:**
 - a. Behind the sulcus terminalis.
 - b. Covered by keratinized stratified squamous epithelium.
 - c. No taste buds.
 - d. von Ebner's glands in the lamina propria.
10. **One of the following is true about the circumvallate tongue papillae:**
 - a. Surrounded by a furrow.
 - b. Scanty taste buds.
 - c. Presence of mucous glands.
 - d. The most numerous papillae.
11. **One of the following is true concerning the foliate papillae:**
 - a. The most numerous papillae in humans.
 - b. Have parallel ridges.
 - c. Have mucous glands in the lamina propria.
 - d. Have scanty taste buds.
12. **The following organ has no taste buds:**
 - a. Soft palate.
 - b. Pharynx.
 - c. Epiglottis.
 - d. Hard palate.
13. **Concerning the taste buds:**
 - a. The taste cells are peripheral.
 - b. Myelinated nerve fibres carry taste sensations.
 - c. Have taste pores.
 - d. Basal cells are sensory cells.
14. **The neuroepithelial taste cells are:**
 - a. Located at the periphery of taste buds.
 - b. Having pale-core vesicles.
 - c. Surrounded by non-myelinated nerve fibres.
 - d. Their vesicles contain digestive enzymes.
15. **The hard palate:**
 - a. Its nasal aspect is lined by respiratory epithelium.
 - b. Its oral aspect is lined by non-keratinized stratified epithelium.
 - c. Its nasal aspect is lined by keratinized stratified epithelium.
 - d. Rich in taste buds.
16. **One of the following is true concerning the Meissner's plexus:**
 - a. It is a sympathetic plexus.
 - b. It supplies the musculosa.
 - c. Present in the mucosa.
 - d. Present in the submucosa.

17. Concerning the Auerbach's plexus:

- a. Between layers of mucosa.
- b. Called the myenteric autonomic plexus.
- c. Supplies the submucosa.
- d. Supplies the muscularis mucosa.

18. The cardiac glands are:

- a. Serous secreting glands.
- b. In the esophageal mucosa.
- c. In the heart.
- d. Enzyme secreting glands.

19. The esophageal glands are:

- a. Endocrine gland.
- b. In the esophageal submucosa.
- c. In the esophageal mucosa.
- d. Their ducts do not open in the lumen.

20. One of the following is true concerning the esophageal musculosa:

- a. Upper 1/3 is mixed muscles.
- b. Middle 1/3 is mixed muscles.
- c. Lower 1/3 is striated muscles.
- d. Middle 1/3 is striated muscles.

21. Concerning the esophagus one of the following is true:

- a. Lined by ker. st. sq. epith.
- b. Has mucosal and submucosal glands.
- c. The musculosa is three layers.
- d. The musculosa is only smooth muscles.

22. Auerbach's plexus is found in:

- a. The muscularis externa.
- b. The submucosa.
- c. The muscularis mucosa.
- d. The mucosa.

23. The fundic glands are characterized by the following:

- a. Occupy 1/2 the thickness of mucosa.
- b. Parallel to the surface.
- c. Simple branched tubular glands.
- d. Abundant C.T. in-between.

24. The fundus of the stomach:

- a. Musculosa is formed of three layers.
- b. Fundic glands are in submucosa.
- c. Adventitia is loose C.T.
- d. Thick fundic sphincter.

25. The fundic glands are lined by:

- a. Oxyphil cells.
- b. Chief cells.
- c. Goblet cells.
- d. Paneth cells.

- 26. The parietal cells are characterized by the following:**
- a. Mainly located in the upper 1/2 of the fundic glands.
 - b. Deeply basophilic.
 - c. Secrete mucus.
 - d. Basal granules.
- 27. One of the following is true concerning the oxyntic cells:**
- a. Secrete rennin.
 - b. Have basal rounded nuclei.
 - c. Apical intracellular canaliculi.
 - d. Apical granules.
- 28. The chief cells are:**
- a. Deeply acidophilic.
 - b. Characterized by intracellular canaliculi.
 - c. Secrete rennin & pepsinogen.
 - d. Have well developed sER.
- 29. The peptic cells are characterized by the following:**
- a. Mainly found in the base of fundic glands.
 - b. Numerous sER.
 - c. Infra-nuclear Golgi apparatus.
 - d. Secrete Hcl.
- 30. The following cell is located in the lining of the fundic mucosa:**
- a. Goblet cells.
 - b. Paneth cells.
 - c. Oxyphil cells.
 - d. Peptic cells.
- 31. One of the following cells is belonging to the gastric APUD cells:**
- a. Secretin cells.
 - b. Oxyphil cells.
 - c. M-cells.
 - d. E-cells.
- 32. Concerning the pyloric glands, one of the following is true:**
- a. Occupy all the thickness of the mucosa.
 - b. Wide and deep gastric pits.
 - c. Perpendicular to surface.
 - d. Crowded.
- 33. The pylorus of the stomach is characterized by:**
- a. Its musculosa is three layers.
 - b. Has scanty C.T. between glands.
 - c. Narrow gastric pits.
 - d. Has thick pyloric sphincter.
- 34. The absorption in the small intestine is augmented by:**
- a. Plica circularis.
 - b. Intestinal musculosa.
 - c. Appendices epiploicae.
 - d. Intestinal serosa.
- 35. The intestinal villi are lined by one of the following cells:**
- a. Goblet cells.
 - b. Peptic cells.
 - c. Mucous neck cells.
 - d. Parietal cells.

36. One of the following is true concerning the Paneth cells:

- a. Line the intestinal villi.
- b. Has apical acidophilic granules.
- c. Secrete intestinal hormones.
- d. Have apical rER.

37. The APUD cells of the intestine are characterized by:

- a. Apical secretory granules.
- b. Basal nuclei.
- c. Infra-nuclear Golgi complex.
- d. Secrete intestinal enzymes.

38. The enterocytes are characterized by:

- a. Line the villi only.
- b. Basal infoldings.
- c. Terminal digestion of fats.
- d. Lateral interdigitations.

39. The microfold cells are:

- a. Found only in the duodenum.
- b. Associated with Brunner's gls.
- c. Antigen-transporting cells.
- d. Specialized cubical cells.

40. The ileum is characterized by:

- a. The villi are broad and long.
- b. Peyer's patches.
- c. Few goblet cells.
- d. Numerous villi.

41. One of the following is true concerning the duodenum:

- a. Few villi.
- b. Contains Peyer's patches.
- c. Numerous villi.
- d. Numerous goblet cells.

42. The colon is characterized by:

- a. Few goblet cells.
- b. Many Paneth cells.
- c. Contains Peyer's patches.
- d. Sub-peritoneal appendices epiploica.

43. The appendix is characterized by:

- a. Projects from the colon.
- b. Irregular villi and crypts.
- c. Appendices epiploica.
- d. Poorly developed muscularis mucosa.

44. The outer longitudinal layer of the musculosa is formed of three bundles in:

- a. Ileum.
- b. Pylorus of stomach.
- c. Jejunum.
- d. Colon.

45. One of the following is true concerning the anal canal:

- a. About 10 cm length.
- b. Has longitudinal folds called the anal sinuses.
- c. Anal columns meet one another to form the pectinate line.
- d. Lined by simple squamous epithelium.

- 46. At the level of the columns of Morgagni, the epithelium is:**
- Simple columnar.
 - Stratified columnar.
 - Simple squamous.
 - Non-keratinized Stratified squamous.
- 47. The rectum differs than the colon in:**
- Deeper and more crypt.
 - Fewer goblet cells.
 - More tenia coli.
 - No peritoneal covering.
- 48. The tenia coli are:**
- Skeletal muscle layer.
 - Continuous inner muscle layer.
 - In the wall of rectum.
 - Three bundles of the outer muscle layer.
- 49. The Peyer's patches are:**
- Present in the rectum.
 - Lymphatic aggregations in the ileum.
 - Piercing the musculosa of the ileum.
 - In the mesenteric side of submucosa.
- 50. The appendices epiploicae are present in:**
- Submucosa of colon.
 - Serosa of appendix.
 - Serosa of colon.
 - Adventitia of rectum.

(B) Problem Solving:

- A 35 old man suffered from anemia, lab investigations proved to be pernicious anemia. This type of anemia may be due to:**
 - Destruction of peptic cells.
 - Destruction of oxyntic cells.
 - Absent APUD cells.
 - Insufficient HCL secretion.
- A 40 years old lady was diagnosed to have peptic ulcer by endoscope. This problem is due to over production of HCL from:**
 - Peptic cells.
 - Oxyphil cells.
 - Mucous neck cells.
 - Parietal cells.
- A middle-aged female accompanied her 7-year old child to the nutrition clinic. History taking revealed that he is losing weight inspite of having regular meals. This case of malabsorption might be due to defect in----- component of enterocytes:**
 - Mitochondria.
 - rER.
 - Cell coat.
 - Cilia.

4. A pregnant female complains of chronic constipation with painful defecation and prolapsed mass from the anus during defecation. Rectal examination revealed hemorrhoids which represent congested:
- Arterial plexus of the anal canal.
 - Submucosal rectal arterial plexus.
 - Mucosal rectal venous plexus.
 - Venous plexus in the submucosa of the anal canal.
5. A male patient was scheduled for performing partial gastrectomy (removal of part of his stomach) after being diagnosed of having cancer stomach. After the operation, he complained of easy fatigability, faintness attacks and headache, suggestive of pernicious anemia. The lost gastric epithelial cells responsible for these symptoms are and are characterized by having:
- Paneth cells / deeply basophilic cytoplasm.
 - Oxyntic cells / pale vacuolated cytoplasm.
 - Parietal cells / deeply acidophilic cytoplasm.
 - Paneth cells / apical secretory granules.

(C) Match Column (A) with Column (B):

Table I

Column (A)	Column (B)
1. The gingiva	a. Cover ant. 2/3 of dorsum of tongue
2. The red margin	b. Covered by thin non-hairy skin
3. The filiform papillae	c. Lined by keratinized. st. sq. epithelium
4. The nasopharynx	d. Lined by non-keratinized st. sq. epithelium
5. The esophagus	e. Lined with respiratory mucosa
	f. Lined with oral mucosa
	g. Covered by thin hairy skin

Table II

Column (A)	Column (B)
1. Peptic cells	a. Antigen transporting cell
2. Paneth cells	b. Basal granules
3. Oxyntic cells	c. In the villi only
4. Microfold cells	d. The most numerous in fundic glands
5. Goblet cells	e. Secrete mucus
	f. In small intestinal crypts only
	g. Deeply acidophilic

Table III

Column (A)	Column (B)
1. APUD cells	a. Secrete mucus
2. Parietal cells	b. Secrete rennin
3. Enterocytes	c. Terminal digestion of carbohydrate
4. Paneth cells	d. Secrete Hcl
5. Chief cells	e. Secrete hormones
	f. Secrete lysozyme
	g. Antigen transporting cells

Answers:

A. MCQ

1	d	13	c	25	b	37	c	49	b
2	c	14	c	26	a	38	d	50	c
3	d	15	a	27	c	39	c		
4	a	16	d	28	c	40	b		
5	c	17	b	29	a	41	c		
6	c	18	b	30	d	42	d		
7	b	19	b	31	d	43	d		
8	c	20	c	32	b	44	d		
9	d	21	b	33	d	45	c		
10	a	22	a	34	a	46	b		
11	b	23	c	35	a	47	d		
12	d	24	a	36	b	48	d		

B. Problem solving

1	b	4	d
2	d	5	c
3	c		

C. Matching

Table I

1	2	3	4	5
c	b	a	e	d

Table II

1	2	3	4	5
d	f	g	a	e

Table III

1	2	3	4	5
e	d	c	F	b

II) Digestive Glands

(A) Multiple Choice Questions (MCQs)

Choose the best answer:

- 1. The major salivary glands are:**
 - a. Compound alveolar exocrine glands.
 - b. Simple branched glands.
 - c. Compound tubular glands.
 - d. Compound tubulo-alveolar glands.

- 2. A major salivary gland:**
 - a. Lingual gland.
 - b. Palatine gland.
 - c. Sublingual gland.
 - d. Von Ebner's gland.

- 3. The mucous cells have:**
 - a. Central rounded nuclei.
 - b. Basophilic cytoplasm.
 - c. Apical acidophilic granules.
 - d. Flat nuclei.

- 4. The serous acini are characterized by:**
 - a. Flat nuclei.
 - b. Vacuolated cytoplasm.
 - c. Few myoepithelial cells.
 - d. Large diameter.

- 5. The serous acini have:**
 - a. Wide lumen.
 - b. Acidophilic cells.
 - c. Tight cell junctions.
 - d. Numerous myoepithelial cells.

- 6. The myoepithelial cells are characterized by:**
 - a. Dome-shaped cell.
 - b. Contain no filaments.
 - c. Allow distension of secretory acini.
 - d. Present between basal lamina and secretory cells.

- 7. The myoepithelial cells are:**
 - a. Few around mucous acini.
 - b. Prevent distension of secretory acini during secretion.
 - c. Present outside the basal lamina.
 - d. Contain actin filaments only.

- 8. The intercalated ducts of the major salivary glands are:**
 - a. Lined with simple columnar cells.
 - b. Their cells are basophilic.
 - c. Found inside the lobules.
 - d. Join to form the main duct.

- 9. Concerning the striated ducts of the glands:**
- Commonly seen in the pancreas.
 - Join to form inter-lobar ducts.
 - Their cells have basal basophilic striations.
 - Have a secretory function.
- 10. The striated ducts:**
- Located in-between the lobules.
 - Have basal infoldings.
 - Acidophilic cells with basal basophilic striations.
 - Lined by pseudostratified columnar epithelium.
- 11. The main ducts of the major salivary glands are lined with:**
- Simple squamous epith.
 - Simple cubical epith.
 - Stratified columnar epith.
 - Pseudo-stratified columnar epith.
- 12. The interlobular ducts of the major salivary glands are lined with:**
- Simple squamous epith.
 - Simple columnar epith.
 - Stratified columnar epith.
 - Pseudo-stratified columnar epith.
- 13. The interlobular ducts of the major salivary glands are:**
- In the C.T. septa between lobes.
 - Have a secretory function.
 - Lined with pseudo-st.col. epith.
 - Lined with simple columnar cells.
- 14. The interlobar ducts of the major salivary glands are lined with:**
- Simple cubical epith.
 - Simple columnar epith.
 - Stratified columnar epith.
 - Pseudo-stratified columnar epith.
- 15. Concerning the parotid gland:**
- Its trabeculae are thin.
 - The parenchyma is almost formed of mucous acini.
 - Stenson's duct opens in the floor of the mouth.
 - The capsule is rich in adipocytes.
- 16. Wharton's duct is the main duct of:**
- Parotid gland.
 - Sublingual gland.
 - Von Ebner's gland.
 - Submandibular gland.
- 17. The parotid gland:**
- Has a thick capsule.
 - Is a seromucous gland in humans.
 - The mucous cells contain lysozyme.
 - Wharton's duct is lined with str. col. epithelium.

18. The sublingual gland is:

- a. Situated in the tongue muscles.
- b. The largest major salivary gland.
- c. A serous gland.
- d. A mucoserous gland.

19. Concerning the liver:

- a. A mixed gland.
- b. A branched tubulo- alveolar gland.
- c. Has thick capsule and trabeculae.
- d. A compound tubulo-alveolar gland.

20. The pancreatic acini are characterized by:

- a. Small diameter with narrow lumen.
- b. The cells are cubical.
- c. Their nuclei are flat.
- d. Has pale vacuolated cytoplasm.

21. The centroacinos cells are:

- a. Flat cells lining the lumen of large ducts.
- b. Columnar cells lining the pancreatic acini.
- c. Myoepithelial cells around the acini.
- d. Telescoped inside the pancreatic acinus.

22. The pancreatic acini are controlled by:

- a. Somatostatin.
- b. Insulin and glucagon.
- c. Cells of Langerhans.
- d. APUD cells of the intestine.

23. The pancreatic main ducts are lined with:

- a. Simple columnar epithelium with goblet cells only.
- b. Stratified columnar epithelium.
- c. Simple columnar epithelium with goblet and APUD cells.
- d. Stratified squamous epithelium.

24. Concerning the liver:

- a. Capsule is fully covered with peritoneum.
- b. Trabeculae are thin delicate in humans.
- c. Lobules are well demarcated in humans.
- d. Liver acinus is a triangular mass of liver tissue.

25. The portal lobule:

- a. is a diamond shaped mass of liver tissue.
- b. Has the portal tract at its center.
- c. Contains parts of two adjacent classic hepatic lobules.
- d. Is divided into three zones.

- 26. The hepatocytes:**
- a. Are binucleated rounded cells.
 - b. Are in direct contact with the endothelial lining of the blood sinusoids.
 - c. Have long microvilli that project into bile canaliculi.
 - d. Have numerous peroxisomes.
- 27. Concerning the hepatic blood sinusoids:**
- a. Lined with fenestrated endothelial cells.
 - b. They are continuous type of capillaries.
 - c. Von Kupffer cells are present in the space of Disse.
 - d. Von Kupffer cells stained with toluidine blue.
- 28. The space of Disse is:**
- a. Located between blood sinusoids and bile canaliculi.
 - b. Lined with fenestrated endothelium.
 - c. Containing Ito cells.
 - d. Called the episinusoidal space.
- 29. One of the following is true concerning the liver:**
- a. The porta hepatis is in its upper surface.
 - b. The largest compound tubular gland.
 - c. A pure endocrine gland.
 - d. The largest compound tubulo-alveolar gland.
- 30. Von Kupffer cells:**
- a. Rich in lysosomes.
 - b. Can be stained with toluidine blue.
 - c. Store vitamin A.
 - d. Present in the space of Disse.
- 31. Ito cells:**
- a. Have numerous lysosomes.
 - b. Present in the space of Disse.
 - c. Phagocytic cells.
 - d. Present in the wall of sinusoids.
- 32. The liver acinus is characterized by:**
- a. A diamond-shaped mass of liver tissue.
 - b. A portal tract is located in its center.
 - c. Parts of three adjacent classic hepatic lobules.
 - d. Between three adjacent central veins.
- 33. Canals of Hering:**
- a. They are spaces between two hepatocytes.
 - b. Microvilli of hepatocytes project into them.
 - c. Sealed from intercellular spaces by tight junctions.
 - d. Lined with simple cubical cells.

34. Concerning the bile canaliculi:

- a. Enclosed between plates of hepatocytes.
- b. Empty in the interlobular ducts.
- c. Bile flows in the same direction of blood.
- d. They are tubular spaces.

35. The canals of Hering are lined by:

- a. Simple squamous epith.
- b. Simple columnar epith.
- c. Stratified columnar epith.
- d. Simple cubical epith.

36. The interlobular bile ducts are lined by:

- a. Simple squamous epith.
- b. Simple columnar epith.
- c. Stratified columnar epith.
- d. Simple cubical epith.

37. One of the following is true about the blood supply of the liver:

- a. 75% oxygen rich blood.
- b. Carried by the portal vein and artery.
- c. 25% portal blood.
- d. Flows from the periphery of the hepatic lobule to the center.

38. The extra hepatic bile ducts are lined by:

- a. Simple squamous epith.
- b. Simple columnar epith.
- c. Stratified columnar epith.
- d. Pseudo-stratified columnar epith.

39. The common hepatic duct is:

- a. Found inside the liver.
- b. Join cystic duct forming the common bile duct.
- c. Lined by stratified columnar epithelium.
- d. Lined by simple squamous epithelium.

40. Sphincter of Oddi is:

- a. Present in cystic duct.
- b. Present in common bile duct.
- c. Regulate passage of bile from common hepatic duct.
- d. Related to the gall bladder.

41. The common bile duct:

- a. Lined by simple columnar epithelium with goblet cells.
- b. Arise from the gall bladder.
- c. Drains bile into the gall bladder.
- d. Formed by the cystic and common hepatic ducts.

42. The gall bladder is lined with:

- a. Transitional epith.
- b. Simple col. epith. with goblet cells.
- c. Simple columnar epith.
- d. Stratified squamous epith.

(B) Problem solving:

1. A 55-years old malnourished woman complain of night blindness 4 months ago. The patient was subsequently diagnosed with vitamin A deficiency. Which of the following cells is related to vitamin A storage?
 - a. Hepatocytes.
 - b. Von Kupffer cells.
 - c. Endothelium.
 - d. Lipocytes.
2. A patient having cancer and treated with chemotherapy. Which cells are affected most?
 - a. Cells in the portal canal.
 - b. Hepatocytes in zone 1.
 - c. Hepatocytes in zone 2.
 - d. Hepatocytes in zone 3.

(C) Match Column (A) with Column (B):

Table I

Column (A)	Column (B)
1. Parotid glands	a. Coiled branched tubular
2. Liver	b. Compound tubulo-alveolar
3. Lingual glands	c. Simple branched tubular
4. Fundic glands	d. Embedded in the tongue muscles
5. Pyloric glands	e. Branched tubulo-alveolar
	f. Compound tubular
	g. Compound alveolar

Table II

Column (A)	Column (B)
1. Striated ducts	a. Stratified columnar epithelium
2. Main salivary duct	b. Cuboidal or low columnar epithelium
3. Main pancreatic duct	c. Flat cells
4. Centroacinosse cells	d. Pseudostratified columnar epithelium
5. Common bile duct	e. Columnar, goblet and APUD cells
	f. Columnar epithelium
	g. Columnar ciliated epithelium

Table III

Column (A)	Column (B)
1. Space of Disse	a. Has three zones
2. Bile canaliculi	b. Lined with flat cells
3. Portal tract	c. Contains Ito cells
4. Liver acinus	d. Has portal tract in the center
5. Portal lobule	e. Tubular spaces between hepatocytes
	f. Between three adjacent hepatic lobules
	g. Lined by Von Kupffer cells

Answers:

A. MCQ

1	d	10	b	19	a	28	c	37	d
2	c	11	c	20	c	29	b	38	b
3	d	12	b	21	d	30	a	39	b
4	a	13	d	22	d	31	b	40	b
5	c	14	d	23	c	32	a	41	d
6	d	15	d	24	b	33	d	42	c
7	b	16	d	25	b	34	d		
8	c	17	a	26	d	35	d		
9	d	18	d	27	a	36	d		

B. Problem solving

1	2
d	b

C. Matching

Table I

1	2	3	4	5
b	f	d	c	a

Table II

1	2	3	4	5
b	a	e	c	f

Table III

1	2	3	4	5
c	e	f	a	d

Urinary System

(A) Multiple Choice Questions (MCQs)

Choose the best answer:

- 1. The glomerular endothelial cells:**
 - a. Are fenestrated without diaphragms.
 - b. Phagocytose old basement membrane.
 - c. Secrete rennin in the filtrate.
 - d. Don't rest on basement membrane.
- 2. Mesangial cells:**
 - a. Phagocytose residues during renewal of glomerular basement membrane.
 - b. Are fenestrated.
 - c. Are present in renal glomerulus only.
 - d. Have primary & secondary processes.
- 3. Concerning macula densa cells:**
 - a. Are called umbrella cells.
 - b. Are tall columnar cells having dark stained nuclei.
 - c. Have basement membrane.
 - d. Are modified cells in wall of proximal convoluted tubules.
- 4. The afferent arteriole:**
 - a. Has thin wall & thick media.
 - b. Has a larger diameter than the efferent arteriole.
 - c. Carries filtered blood.
 - d. Has thin wall & thin media.
- 5. Renal column of Bertin is:**
 - a. Finger like extension of medullary tissue that enter the cortex.
 - b. Cortical tissue between renal pyramids.
 - c. Renal pyramid & associated cortical tissue.
 - d. Cells present between efferent & afferent arterioles.
- 6. Renal labyrinth is the:**
 - a. Cortical tissue extending between medullary pyramids.
 - b. Expanded end of ureter.
 - c. Radial striations projecting from medullary pyramids.
 - d. Cortical tissue surrounding a medullary ray.

7. Renal lobe is:

- a. Medullary ray surrounding cortical tissue.
- b. A renal pyramid with its surrounding cortical tissue.
- c. Cortical tissue surrounding a medullary ray.
- d. Cortical tissue extending between medullary pyramids.

8. Medullary ray is:

- a. Cortical tissue under capsule.
- b. Cortical tissue extends between medullary pyramids.
- c. Medullary pyramid & overlying cortical tissue.
- d. Parallel straight tubules of loop of Henle and collecting tubules in the cortex.

9. Podocyte:

- a. Major but not minor processes project from cell body.
- b. Line the parietal layer of Bowman's capsule.
- c. Separated from basement membrane by capsular space.
- d. Filtration slits are not covered by diaphragms.

10. Glomerular capillaries are:

- a. Lined with non-fenestrated endothelium.
- b. The basement membrane is thin.
- c. With EM, the basement membrane is formed of outer & inner electron dense & central electron lucent layer.
- d. Endothelium allows some plasma proteins to pass.

11. The blood renal barrier is formed of:

- a. Continuous endothelium of efferent arteriole.
- b. Cuboidal cells in proximal convoluted tubule.
- c. Mesangial cells.
- d. Continuous glomerular basement membrane.

12. Mesangial cells:

- a. Share in formation of blood renal barrier.
- b. Line the visceral layer of Bowman's capsule.
- c. Have phagocytic function.
- d. Secrete medullipin hormone.

13. Distal convoluted tubules:

- a. Are 3-5 cells in cross section.
- b. Cytoplasm is deeply acidophilic.
- c. Have clear cell boundaries & less apparent brush border.
- d. Is 60µm in diameter.

14. The loop of henle:

- a. Has thin descending limb similar to proximal convoluted tubule.
- b. Has thick ascending limb lined by simple squamous epithelium.
- c. Influence the concentration of urine.
- d. Cortical nephrons have very long loops.

15. The juxtaglomerular cells are formed of:

- a. Podocytes.
- b. Modified smooth muscles mainly in media of afferent arteriole.
- c. Intraglomerular mesangial cells.
- d. Columnar cells in wall of collecting tubules.

16. The prostatic urethra is lined by:

- a. Pseudostratified columnar epith.
- b. Stratified columnar epith.
- c. Stratified squamous epith.
- d. Transitional epith.

17. The membranous urethra is lined by:

- a. Pseudostratified columnar epith.
- b. Simple squamous epith.
- c. Stratified squamous epith.
- d. Transitional epith.

18. The fossa navicularis of male urethra is lined by:

- a. Pseudostratified columnar epith.
- b. Stratified columnar epith.
- c. Stratified squamous epith.
- d. Transitional epith.

19. The penile urethra above fossa navicularis is lined by:

- a. Pseudostratified columnar epith.
- b. Stratified columnar epith.
- c. Stratified squamous epith.
- d. Transitional epith.

20. The female urethra is:

- a. Lined by transitional epithelium & stratified columnar epithelium.
- b. A curved tube.
- c. 20-25 cm long.
- d. Pathway for semen & urine.

21. Macula densa cells:

- a. Contain rennin granules in cytoplasm.
- b. Are called epitheloid cells.
- c. Have phagocytic function.
- d. Golgi apparatus is present in basal region.

22. The juxta glomerular cells:

- a. Have closely packed nuclei.
- b. Have phagocytic function.
- c. Have golgi apparatus present in basal region.
- d. Have rennin granules in cytoplasm.

- 23. Extra glomerular mesangial cells:**
- a. Contain rennin granules in cytoplasm.
 - b. Are present between the glomerular capillaries.
 - c. Show basal infoldings.
 - d. Are present between afferent, efferent arterioles & macula densa cells.
- 24. Intra glomerular mesangial cells:**
- a. Contain rennin granules in cytoplasm.
 - b. Show apical brush border.
 - c. Have Golgi apparatus present in basal region.
 - d. Cytoplasm is rich in lysosomes.
- 25. In diabetic nephropathy the most affected part is:**
- a. Proximal convoluted tubule.
 - b. The glomerular barrier.
 - c. The distal convoluted tubule.
 - d. The urethra.
- 26. Diabetes insipidus results from insufficient production of:**
- a. Medullipin i.
 - b. Medullipin ii.
 - c. ADH.
 - d. Aldosterone.
- 27. Loss of voluntary control over the external sphincter muscle of urethra leads to:**
- a. Urine incontinence.
 - b. Hematuria.
 - c. Albuminuria.
 - d. Diabetes insipidus.
- 28. medullary collecting tubules are:**
- a. Present in the medullary rays.
 - b. Lined by intercalated cells only.
 - c. Formed by the union of several cortical collecting tubules.
 - d. Open into the tips of renal papillae.
- 29. Ducts of bellini are characterized by:**
- a. Highly convoluted collecting tubules.
 - b. The major factor in urine concentration.
 - c. Not affected by ADH.
 - d. Lined with simple squamous epithelium.
- 30. Principal cells lining the renal collecting tubules are:**
- a. Two types.
 - b. Dark cells.
 - c. Each cell has a single cilium.
 - d. Ill-defined cell boundaries.
- 31. Intercalated cells of collecting tubules are characterized by:**
- a. Have two types.
 - b. Have basal infoldings.
 - c. Increase water permeability of collecting tubules.
 - d. Pale cells.

32. In case of acidemia:

- a. There is predominance of principal cells.
- b. Intercalated cells secrete excess HCO_3 in urine.
- c. Intercalated cells secrete excess HCO_3 in urine.
- d. Intercalated cells secrete excess H^+ & generate more HCO_3 .

33. In cases of alkalemia:

- a. There is predominance of principal cells.
- b. Intercalated cells secrete excess HCO_3 in urine.
- c. Intercalated cells secrete excess HCO_3 in urine.
- d. Intercalated cells secrete excess H^+ & generate more HCO_3 .

34. Podocytes are:

- a. Cells lining parietal layer of bowman's capsule.
- b. Glomerular endothelial cells.
- c. Cells lining visceral layer of bowman's capsule.
- d. Lacis cells.

(B) Problem solving:

1. **A 40 years old man with normal blood sugar level complaining of excessive production of diluted hypotonic pale urine. His complain may be due to:**
 - a. Excessive production of ADH.
 - b. Inadequate reabsorption of water from proximal convoluted tubule.
 - c. Affection of renin-angiotensin-aldosterone system.
 - d. Inadequate reabsorption of water from collecting tubule.
2. **A 30 years old lady diagnosed clinically as having diabetic nephropathy. On urine analysis, RBCs and albumin were detected. This condition may result from:**
 - a. The renin-angiotensin-aldosterone system is affected.
 - b. The juxtaglomerular apparatus is disrupted.
 - c. The renal glomeruli are affected and the renal barrier is disrupted.
 - d. The internal and external mesangial cells are affected.
3. **A 30 years patient suffers from chronic essential hypertension (180/110), blood test revealed hypernatremia and hypokalemia. This could be attributed to hypersecretion of:**
 - a. Lacis cells.
 - b. Macula densa.
 - c. Podocyte.
 - d. Juxtaglomerular cells.
4. **A 40 years old woman complaining of urinary incontinence , this may result from lesion of:-**
 - a. Smooth muscle fibers of external urethral sphincter.
 - b. Skeletal muscle fibers of internal urethral sphincter.
 - c. Smooth muscle fibers of internal urethral sphincter.
 - d. Skeletal muscle fibers of external urethral sphincter.

(C) Match Column (A) with Column (B):

Table I

Column (A)	Column (B)
1. Prostatic part of male urethra	a. Short straight tube
2. Membranous part of male urethra	b. Curved tube about 20 cm
3. Urinary bladder	c. Lined by pseudostratified columnar epithelium
4. Ureter	d. Prominent longitudinal folds giving lumen stellate appearance
5. Penile urethra	e. The initial part of urethra of male urethra
	f. The transitional epithelium is 5-6 layers in empty state
	g. The glands of littre open into it

Table II

Column (A)	Column (B)
1. Proximal convoluted T.	a. Each 6-8 of them open into duct of Bellini
2. Distal convoluted tubule	b. Lined by simple squamous epithelium
3. Medullary collecting tubule	c. Allow substances less than 400,000 Dalton
4. Thin ascending & descending Loop of Henle	d. Allow the passage of small molecules less than 40,000 Dalton
5. Glomerular basement membrane	e. Deeply a acidophilic cytoplasm & basal striations
	f. Lined by 5-8 cubical cells, less apparent brush border
	g. Prevents passage of large molecules more than 160,000

Table III

Column (A)	Column (B)
1. Podocytes	a. It allow passage of anything smaller than red cells & blood platelets
2. Mesangial cells	b. Modified smooth muscle in wall of afferent arteriole
3. Macula densa	c. Line parietal layer of Bowman'scapsule
4. Juxtaglomerular cells	d. Support glomerular basement membrane & secrete extracellular matrix
5. Fenestrated endothelium	e. Have basal striations
	f. Columnar cells present in the distal convoluted tubule, with absent basement membrane
	g. Separated from basement membrane by subpodocytic space

Answers

A. MCQ

1	c	10	a	19	c	28	a
2	a	11	d	20	b	29	c
3	a	12	d	21	a	30	b
4	b	13	c	22	d	31	c
5	b	14	c	23	d	32	a
6	b	15	c	24	d	33	d
7	d	16	b	25	d	34	b
8	b	17	d	26	b		
9	d	18	a	27	c		

B. Problem Solving:

1	d
2	c
3	d
4	d

C. Matching

Table I

1	2	3	4	5
e	c	f	d	g

Table II

1	2	3	4	5
e	f	a	b	g

Table III

1	2	3	4	5
g	d	f	b	a

Endocrine Glands

(A) Multiple Choice Questions (MCQs)

Choose the best answer:

- 1. Which of the following produces ACTH:**
 - a. basophils of pars distalis.
 - b. acidophils of pars distalis.
 - c. chromophobes of pars distalis.
 - d. neurosecretory cells of the hypothalamus.
- 2. Growth hormone:**
 - a. is secreted by the acidophils of pars distalis.
 - b. its secretion is stimulated by somatostatin.
 - c. its secretion is suppressed by G.H releasing hormone.
 - d. increased secretion causes dwarfism.
- 3. Orange G specifically stains**
 - a. chromophobes of pars distalis.
 - b. pituicytes of pars nervosa.
 - c. basophils of pars distalis.
 - d. somatotrophs of pars distalis.
- 4. ACTH stimulates the production of**
 - a. cortisol.
 - b. androgen from the zona glomerulosa.
 - c. aldosterone.
 - d. adrenaline from chromaffin cells.
- 5. The adrenocorticotrophic hormone is secreted from:**
 - a. neurohypophysis.
 - b. adenohypophysis.
 - c. adrenal cortex.
 - d. pars nervosa.
- 6. Mammotrophs:**
 - a. secrete growth hormone.
 - b. decrease in number during pregnancy.
 - c. stain with orange G.
 - d. enlarge during pregnancy.
- 7. Corticotrophs secrete:**
 - a. ACTH.
 - b. TSH.
 - c. Follicle stimulating hormone.
 - d. Oxytocin.
- 8. The secretion of which one is controlled by hypothalamo-hypophyseal feedback?**
 - a. thyroxin.
 - b. aldosterone.
 - c. insulin.
 - d. calcitonin.

9. **Basophils are:**
 - a. smaller than acidophils in diameter.
 - b. less than acidophils in number.
 - c. positive for Orange G stain.
 - d. have numerous cisternae of sER.
10. **In reptiles, pars intermedia secretes:**
 - a. ACTH.
 - b. TSH.
 - c. MSH.
 - d. FSH.
11. **ACTH- is produced by**
 - a. basophils of the pars intermedia.
 - b. acidophils of the pars distalis.
 - c. basophils of the pars distalis.
 - d. neurosecretory cells of the hypothalamus.
12. **Concerning the pars nervosa, it:**
 - a. can appear as brain sand.
 - b. contains nerve cells.
 - c. has axons of hypothalamo-hypophyseal tract.
 - d. has portal circulation.
13. **Pituicytes are found in:**
 - a. pars distalis.
 - b. pars intermedia.
 - c. pars nervosa.
 - d. pars tuberalis.
14. **The posterior pituitary:**
 - a. contains pituicytes, herring bodies and axons.
 - b. has a portal blood supply.
 - c. contains nerve cells.
 - d. form oxytocin & vasopressin.
15. **Pars nervosa contains:**
 - a. nerve fibres.
 - b. basophils.
 - c. oxyphils.
 - d. C – cells.
16. **Pituicytes:**
 - a. produce vasopressin.
 - b. produce prolactin.
 - c. a type of glial cell.
 - d. found in pars distalis.
17. **The hypothalamus is connected with pars nervosa via**
 - a. hypothalamo-hypophyseal tract.
 - b. pars intermedia.
 - c. pars tuberalis.
 - d. supraoptic & paraventricular nuclei.
18. **Oxytocin is synthesized in:**
 - a. pars nervosa.
 - b. adenohypophysis.
 - c. hypothalamus.
 - d. adrenal medulla.

- 19. The releasing and inhibiting hormones are transported to pars distalis via:**
- a. superior hypophyseal arteries.
 - b. portal vessels.
 - c. inferior hypophyseal vein.
 - d. inferior hypophyseal artery.
- 20. Vasopressin & oxytocin are synthesized in the:**
- a. pars distalis.
 - b. pars intermedia.
 - c. pars nervosa.
 - d. supraoptic & paraventricular nuclei.
- 21. Mineralocorticoids:**
- a. are regulated by angiotensin II.
 - b. are protein in nature.
 - c. are secreted from zona fasciculata.
 - d. regulate carbohydrate metabolism.
- 22. Which of the following cells have no granules:**
- a. acidophils of pars distalis.
 - b. basophils of pars distalis.
 - c. cells of adrenal cortex.
 - d. chromaffin cells.
- 23. The main ultrastructural feature of the zona fasciculata cells is the presence of:**
- a. smooth endoplasmic reticulum.
 - b. rough endoplasmic reticulum.
 - c. few mitochondria.
 - d. long cilia.
- 24. Chromaffin cells of:**
- a. pars distalis produce growth hormone.
 - b. zona fasciculata produce androgens.
 - c. suprarenal medulla produce epinephrine.
 - d. zona glomerulosa produce mineralocorticoids.
- 25. The following statement is true about thyroid follicles:**
- a. Lined with squamous cells.
 - b. Form T_3 , T_4 & TSH.
 - c. Surrounded by fenestrated capillaries.
 - d. Lined with parafollicular cells which store calcitonin in the colloid.
- 26. The adrenal cortex secretes:**
- a. calcitonin.
 - b. mineralocorticoids.
 - c. catecholamines.
 - d. ACTH.
- 27. The cells of the adrenal cortex:**
- a. are ectodermal.
 - b. are rich in rough endoplasmic reticulum.
 - c. produce steroid hormones.
 - d. produce noradrenaline.

- 38. With EM the characteristic feature of zona fasciculata is:**
- a. lipid droplets.
 - b. lysosomes.
 - c. rER.
 - d. electron dense granules.
- 39. Adrenocortical cells have:**
- a. Few mitochondria.
 - b. large amount of rER.
 - c. abundant SER.
 - d. steroid hormone-containing secretory granules.
- 40. Thyroid follicles:**
- a. are usually lined by a simple squamous epithelium.
 - b. contain follicular cells that release TSH.
 - c. contain parafollicular cells that release calcitonin.
 - d. have basophilic colloid in the lumen.
- 41. Thyroglobulin is stored in:**
- a. the lumen of the follicles.
 - b. Golgi apparatus.
 - c. lysosomes.
 - d. mitochondria.
- 42. In follicular cells of the thyroid gland lysosomes are essential for:**
- a. iodination of thyroglobulin.
 - b. Synthesis of thyroglobulin.
 - c. hydrolysis of thyroglobulin into $T_3 + T_4$.
 - d. release of calcitonin.
- 43. The colloid in thyroid follicles:**
- a. is seen only by EM.
 - b. usually is basophilic.
 - c. contains calcitonin.
 - d. contains thyroglobulin.
- 44. The main ultrastructural features of the thyroid follicular cell is presence of:**
- a. rough and smooth endoplasmic reticulum.
 - b. rough endoplasmic reticulum and lysosomes.
 - c. few mitochondria.
 - d. basal secretory vesicles.
- 45. Thyroid interfollicular cells secrete:**
- a. parathyroid hormone.
 - b. thyroxin hormone.
 - c. calcitonin hormone.
 - d. calcium.
- 46. Iodine is essential for the synthesis of:**
- a. parathormone.
 - b. calcitonin.
 - c. thyroxin.
 - d. cortisol.

47. **Normal parathyroid glands of adults possess:**
- a. oxyphil cells.
 - b. chromaffin cells.
 - c. basophil cells.
 - d. parafollicular cells.
48. **Parafollicular cells:**
- a. have typical steroid cell characters.
 - b. are involved in the uptake of thyroglobin.
 - c. present between the basement membrane and follicular cells.
 - d. respond to thyroid stimulating hormone (TSH).
49. **Oxyphil cells of the parathyroid gland are characterized by**
- a. increase in number with age.
 - b. rich in rER & Golgi apparatus.
 - c. they represent degenerated chief cells.
 - d. secrete parathyroid hormone.
50. **Which is true about oxyphil cells:**
- a. are deeply acidophilic.
 - b. contain numerous rER
 - c. are more numerous than chief cells.
 - d. smaller than chief cells.
51. **APUD cells:**
- a. Are of six types.
 - b. Include cells of suprarenal cortex.
 - c. Include renal juxtaglomerular cells.
 - d. Are deeply acidophilic.
52. **Oxyphil cells are present in the**
- a. parathyroid gland.
 - b. adenohypophysis.
 - c. neurohypophysis.
 - d. pineal gland.
53. **Oxyphil cells of the parathyroid gland are acidophilic due to the presence of many:**
- a. sER.
 - b. rER.
 - c. lysosomes.
 - d. mitochondria.
54. **Iodination of the thyroglobulin occurs:**
- a. extracellular in the follicular lumen.
 - b. intracellular inside the follicular cell.
 - c. intercellular.
 - d. inside the capillary bed.

- 55. The secretion of parathyroid cells is regulated by:**
 a. Thyroid stimulating hormone. b. Blood calcium level.
 c. Corticosteroid level. d. T_3 & T_4 .
- 56. Pinealocytes of the pineal body:**
 a. produce melatonin only. b. are astrocyte-like cells.
 c. are called brain sand. d. produce serotonin & melatonin.
- 57. The pineal body:**
 a. attached to 4th ventricle by pineal stalk.
 b. has a capsule formed of the dura matter.
 c. contain calcified concretion called brain sand.
 d. secrete cortisone.
- 58. Melatonin is secreted from:**
 a. chromaffin cells. b. astroglia cells. c. corticotrophs. d. pinealocytes.
- 59. Pinealocytes:**
 a. secrete melatonin only. b. secrete serotonin only.
 c. appear as brain sand. d. secrete melatonin and serotonin.
- 60. Brain sand:**
 a. is deeply basophilic.
 b. serves as a landmark in x-ray.
 c. is found in the diseased pineal body.
 d. decrease in size with age.

(B) Problem solving:

- 1. A patient introduced to outpatient clinic complaining of severe headache and nausea. On examination, there was broadening of all his bones and enlargement of the face bones (acromegaly). MRI (advanced radiologic investigation) revealed a functioning tumor affecting the pituitary gland. The tumor originates from the following type of cells:**
 a. Mammothrophs. b. LH. secreting cells.
 c. FSH secreting cells. d. Somatotrophs.
- 2. A patient came to the emergency room complaining of a very painful sustained contraction of most of his muscle (tetany). Investigations revealed a low blood calcium level in spite that he takes a great supplement of calcium. This may be due to deficiency of:**
 a. Adrenaline. b. thyrocalcitonin.
 c. cortisone. d. parathormone.

3. A young man, 18 years, came with his parents. They complain of his short stature, 80 cm tall, in spite of giving him a good nutrition rich in all food elements with a lot of vitamins and calcium. Laboratory investigations may reveal a low level of:
 - a. ACTH.
 - b. cortisone.
 - c. growth hormone.
 - d. adrenaline.
4. A young female complained that the pregnancy did not occur after 4 years of marriage. Investigations revealed that there was no ova production of both ovaries. Laboratory investigations revealed very low serum level of:
 - a. ACTH
 - b. growth hormone
 - c. luteinizing hormone
 - d. FSH
5. A patient was carried to the hospital with severe renal colic. The investigations revealed many renal stones, formed mainly of calcium salts, and a very high serum calcium level. This condition may result of hyper function of:
 - a. parathyroid gland.
 - b. thyroid gland.
 - c. basophils of pituitary gland.
 - d. suprarenal medulla.
6. A 40 year old man complained of arrhythmias and unexplained anxiety, upon examination, life threatening hypertension was recorded. This case is likely diagnosed as:
 - a. Iodine deficiency goiter.
 - b. Hypoparathyroidism.
 - c. Pheochromocytoma.
 - d. Cushing syndrome.
7. In your endocrinology department, five patients were diagnosed to have:
 - a. Deficient T_3 and T_4 production.
 - b. Osteoporosis.
 - c. Tetany.
 - d. Gigantism.

Which one is suffering from pituitary adenoma?
8. After thyroid surgical procedure, a middle-aged woman came to the emergency room presenting with spastic contraction of skeletal muscles. This case should be treated with:
 - a. Calcitonin supplementation.
 - b. Calcium supplementation.
 - c. Thyroid hormone supplementation.
 - d. Muscle relaxant.
9. A 20 years old woman presenting with a huge swelling in the front of the neck came to the clinic, after taking the patient's history, you discovered that she lives in mountainous region where food is grown in iodine-poor soil. A possible diagnosis to this case could be:
 - a. Pituitary adenoma.
 - b. Endemic goiter.
 - c. Thyrotoxicosis.
 - d. Addison's disease.

10. Ahmad, a 25 year old healthy man was on a jet flight from Cairo, Egypt to California, USA. The trip extended for 35 hours including two stops for 9 hours. When Ahmad arrived at the hotel he couldn't sleep and felt light-headed, irritated and had a hard time focusing. Which of the following structures is responsible for alleviating these symptoms?

- a. Carotid body.
- b. Herring body.
- c. Pineal body.
- d. Basal body.

(C) Match Column (A) with Column (B):

Table I

Column (A)	Column (B)
1. T3, T4	a. Pinealocytes
2. Melatonin	b. Spongocytes
3. Cortisol	c. Parafollicular cell
4. Calcitonin	d. Follicular cells of thyroid
5. Adrenaline	e. Corticotrophs
	f. Chromaffin cells
	g. Thyrotrophs

Table II

Column (A)	Column (B)
1. Serotonin	a. Follicular cells
2. Carminophils	b. Liver
3. Prolactin	c. APUD
4. Oxytocin	d. Mammatrophs
5. MSH	e. pars intermedia cells
	f. Milk secretion
	g. Milk ejection

Table III

Column (A)	Column (B)
1. Pituicytes	a. Pars nervosa
2. Herring bodies	b. Corticotrophs
3. Lipotropic factor	c. Neuroglial cells
4. Zona glomerulosa	d. Growth hormone
5. APUD cells	e. Aldosterone
	f. C-cells
	g. Thyrotrophs

Table IV

Column (A)	Column (B)
1. Chief cells	a. Cortisol
2. Follicular cells	b. Calcitonin
3. Spongiocytes	c. Parathyroid hormone
4. Parafollicular cells	d. T3& T4
5. Brain sand	e. Mineralocorticoids
	f. Androgen
	g. Pineal body

Table V

Column (A)	Column (B)
1. Zona fasciculata	a. Pregnancy cells
2. Paranglia	b. Castration cells
3. Gonadotrophs	c. Spongiocytes
4. Mammotrophs	d. Chromaffin cells
5. Zona reticularis	e. Brain sand
	f. Androgen
	g. Thyroid hormone

Table VI

Column (A)	Column (B)
1. Hormone producing tumor	a. Bone cavities
2. Pheochromocytoma	b. Tetany
3. Hyperparathyroidism	c. C – cells
4. Hypoparathyroidism	d. Tumor of adrenal medulla cells
5. APUD	e. Immunocytochemical identification
	f. Pars nervosa
	g. Orange G stain

Answers

A. MCQ

1	a	13	c	25	c	37	d	49	a
2	a	14	a	26	b	38	a	50	a
3	d	15	a	27	c	39	c	51	c
4	a	16	c	28	a	40	c	52	a
5	b	17	a	29	b	41	a	53	d
6	d	18	c	30	c	42	c	54	a

7	a	19	b	31	a	43	d	55	b
8	a	20	d	32	d	44	b	56	d
9	b	21	a	33	c	45	b	57	c
10	c	22	c	34	b	46	c	58	d
11	c	23	a	35	d	47	a	59	d
12	c	24	c	36	c	48	c	60	b

B. Problem Solving:

1	2	3	4	5	6	7	8	9	10
d	d	c	d	a	c	d	b	b	c

C. Matching

Table I

1	2	3	4	5
d	a	b	c	f

Table II

1	2	3	4	5
c	d	f	g	e

Table III

1	2	3	4	5
c	a	b	e	f

Table IV

1	2	3	4	5
c	d	a	b	g

Table V

1	2	3	4	5
c	d	b	a	f

Table VI

1	2	3	4	5
e	d	a	b	c

The Male Genital System

(A) Multiple Choice Questions (MCQs)

Choose the best answer:

1. **The testis is a:**
 - a. Compound tubular gland.
 - b. Simple branched tubular gland.
 - c. Compound tubuloalveolar gland.
 - d. Simple coiled tubular gland.
2. **The mediastinum testis is the posterior thickening of the:**
 - a. Tunica vasculosa.
 - b. Scrotum.
 - c. Tunica albuginea.
 - d. Tunica vaginalis.
3. **Spermatogonia are characterized by:**
 - a. Containing the haploid number of chromosomes.
 - b. Lying in the luminal compartment of the seminiferous tubules.
 - c. Being of three types.
 - d. Lying in one layer only.
4. **Primary spermatocytes:**
 - a. Small cells with dark rounded nuclei.
 - b. Have characteristic appearances in testicular sections.
 - c. Changed to secondary spermatocyte by mitotic cell division.
 - d. Present in the basal compartment.
5. **The following occurs during spermiogenesis:**
 - a. The anterior 2/3 of the sperm nucleus is covered by the acrosome.
 - b. Hydrolytic enzymes are formed in sER.
 - c. Mitochondria are not apparent in the sperm cell during this process.
 - d. The formed residual bodies are phagocytosed by macrophages.
6. **Spermatogenesis in man lasts for:**
 - a. 84 days
 - b. 74 days
 - c. 64 days
 - d. 54 days
7. **Spermiogenesis is the process by which:**
 - a. Spermatogonia differentiate into spermatids.
 - b. Spermatids change into spermatozoa by metamorphosis.
 - c. Spermatogonia migrate to the primitive male gonads.
 - d. Spermatozoa are propelled towards the duct system.
8. **The following characterizes acrosomal phase of spermiogenesis:**
 - a. Migration of two centrioles.
 - b. Pinching off excess cytoplasm.
 - c. Acrosomal vesicle covers anterior 1/3 of nucleus.
 - d. Mitochondria are arranged to form the middle piece segment.

9. **Spermiation failure:**
- Could result from gonadotropin suppression.
 - Sperms have inefficient movement.
 - Retained sperms are phagocytosed by macrophages.
 - It is the failure of testicular descent to the scrotum.
10. **During the Golgi phase of spermiogenesis:**
- Mitochondria aggregate below the neck.
 - Acrosomal cap spreads to cover the anterior 2/3 of nucleus.
 - Two centrioles migrate to the opposite pole of the spermatid.
 - Excess cytoplasm is pinched off.
11. **The following hormone suppresses spermatogenesis:**
- Inhibin.
 - Testosterone.
 - Anti-Mullerian hormone.
 - FSH.
12. **Spermatogenic cell having haploid number of chromosome is the:**
- Secondary spermatocyte.
 - Primary spermatocyte.
 - Dark type A spermatogonia.
 - Type B spermatogonia.
13. **Secondary spermatocytes:**
- Contain diploid number of chromosomes.
 - Produced under direct effect of LH.
 - Difficult to be seen in testicular sections.
 - Is the final step in spermatogonial stage.
14. **The following characterizes androgen binding protein of the testis:**
- It is produced by Leydig cells.
 - It is produced under the control of LH of anterior pituitary.
 - It accumulates testosterone inside the seminiferous tubule.
 - It is secreted in the pre-pubertal testis.
15. **The following characterizes Sertoli cells:**
- They have large dark nuclei.
 - They are sensitive to harmful substances.
 - They have well-defined lateral borders.
 - They have pale acidophilic cytoplasm.
16. **In EM sections, the following is a feature of Sertoli cells:**
- Scanty rER & sER.
 - Numerous mitochondria and lysosomes.
 - Poorly developed cytoskeleton of microtubules.
 - Few lipid droplets.

17. **The Interstitial cells of Leydig:**
- a. Easily affected by elevated testicular temperature.
 - b. Show lipofuscin granules in elderly.
 - c. Very rich in rER.
 - d. Have basophilic cytoplasm.
18. **LM picture of Leydig cells shows the following:**
- a. Cells forming acini.
 - b. Reinke's crystalloids within their cytoplasm.
 - c. Branched cells with dark nuclei.
 - d. Polygonal cells among fenestrated blood capillaries.
19. **Sperm maturation:**
- a. Is achieved in the efferent ductules.
 - b. Is due to glycoproteins secreted by the epididymis.
 - c. Means that sperms are fully capacitated.
 - d. Starts only in female fallopian tube.
20. **The male intratesticular ducts include:**
- a. Ejaculatory ducts.
 - b. Rete testis.
 - c. Epididymis.
 - d. Vas deferens.
21. **Concerning the epididymis:**
- a. It is a single tubule 45 cm long.
 - b. It is lined by pseudostratified columnar ciliated epithelium (motile cilia).
 - c. It is responsible for storage and maturation of spermatozoa.
 - d. It secretes testicular fluid.
22. **Stereocilia are found in:**
- a. Rete testis.
 - b. Seminal vesicle.
 - c. Epididymis.
 - d. Vasa efferentia.
23. **Concerning the vas deferens the following statement is correct:**
- a. It is a single muscular tube 4-6 m long.
 - b. Its ampulla is joined by duct of seminal vesicle to form ejaculatory duct.
 - c. Lined by stratified columnar epithelium.
 - d. Its musculosa is formed of two layers of smooth muscle fibers.
24. **Regarding benign prostatic hyperplasia:**
- a. Flow of urine is not affected.
 - b. Prostatic mucosal and submucosal acini are enlarged.
 - c. Affects mainly the main prostatic acini.
 - d. More common in young age.

25. The following is not among the content of spermatic cord:
- a. Vas deferens.
 - b. Dartos muscle.
 - c. Pampiniform plexus of veins.
 - d. Testicular artery.
26. The seminal vesicles:
- a. Lined by stratified columnar epithelium.
 - b. Their secretion gives about 25% of the seminal fluid.
 - c. Their secretion is viscid & yellowish.
 - d. Their secretion is rich in fibrinolysin.
27. The prostate is a:
- a. Compound tubular gland.
 - b. Compound tubuloalveolar gland.
 - c. Simple branched tubular gland.
 - d. Compound alveolar gland.
28. Bulbo-urethral glands:
- a. Surround prostatic urethra.
 - b. Secrete clear serous fluid.
 - c. Its secretion is rich in abundant sugars.
 - d. Lined by pseudostratified columnar epithelium.
29. The duct of Cowper's gland opens into:
- a. Prostatic urethra.
 - b. Membranous urethra.
 - c. Penile urethra.
 - d. Ejaculatory duct.
30. The interconnecting vascular spaces in the penis are separated by trabeculae made of:
- a. Collagen fibers and skeletal muscles.
 - b. Reticular and elastic fibers.
 - c. Collagen, reticular and elastic fibers.
 - d. Smooth muscles, collagen and elastic fibers.

(B) Problem solving:

1. A male infant was examined shortly after birth. His scrotal bag was empty. He was diagnosed as cryptorchidism. If left untreated, later he would suffer:
- a. Failure of development of secondary sex characters only.
 - b. Immotile cilia syndrome.
 - c. Infertility with normally developed secondary sex characters.
 - d. Infertility with undeveloped secondary sex characters.

2. **A 32-year-old male complained from pain on right side of scrotum, sense of heaviness on the same side. With ultrasonographic examination, the diagnosis was varicocele. This may cause:**
- Hyperthermia of testis affecting only quantity of sperm.
 - Hypothermia of testis affecting only quality of sperm.
 - Hyperthermia of testis affecting both quality and quantity of sperm.
 - Hypothermia of testis affecting both quality and quantity of sperm.
3. **A 60 years old male complaining from difficulty in micturition especially at night. Sometimes urine is retained and he fails to evacuate his bladder. He was diagnosed to have benign prostatic hyperplasia. This might result from:**
- Enlargement of main and submucosal acini of prostate.
 - Enlargement of submucosal and mucosal acini of prostate.
 - Enlargement of transitional epithelium lining the prostatic urethra.
 - Enlargement of main and mucosal acini of prostate.
4. **A - 35 years-old married male was infertile. He also suffered from chronic respiratory tract infections. He was diagnosed to have primary ciliary dyskinesia, which might result from:**
- Lack of proteins needed for microvilli elongation and shortening.
 - Lack of proteins needed for muscle contractions.
 - Lack of proteins needed for mucous formation.
 - Lack of proteins needed for ciliary and flagellar motility.
5. **A 65-year-old man was diagnosed clinically as suffering from prostatic carcinoma. Which of the following statements concerning his case is correct?**
- Difficult micturition could be his early complaint.
 - Mucosal and submucosal prostatic acini are enlarged.
 - Main group of the prostatic acini is enlarged.
 - He has very low blood levels of prostatic surface antigen (PSA).
6. **A 62-year-old man with markedly elevated levels of circulating PSA. This patient may be suffering from:**
- Prostatic cancer only.
 - Benign prostatic hyperplasia.
 - Prostatic cancer or prostatic inflammation.
 - Prostatic inflammation only.
7. **A 35-year-old man has complete absence of spermatozoa on repeated semen analysis, however his testicular biopsy revealed many spermatozoa inside the seminiferous tubules. This condition could reflect:**
- Failure of the Sertoli cells to secrete the testicular fluid.
 - Low levels of testosterone secreted from Leydig's cells.
 - Presence of anti-sperm antibodies in his blood.
 - Bilateral congenital absence of the vas deferens.

(C) Match Column (A) with Column (B):

Table I

Column (A)	Column (B)
1. Leydig cells	a. Largest cells in spermatogenic series
2. Sertoli cells	b. Undergo metamorphosis
3. Spermatids	c. Secrete testosterone
4. Secondary spermatocytes	d. Give primary spermatocytes
5. Type B spermatogonia	e. Secrete testicular fluid
	f. Undergo 2nd meiotic division
	g. Have head & tail regions

Table II

Column (A)	Column (B)
1. Seminal vesicle	a. Mucus-secreting gland
2. Prostate	b. Its secretion is rich in fructose
3. Spermatid cord	c. Dartos muscle
4. Epididymis	d. Storage & maturation of sperms
5. Cowper's gland	e. Its secretion is rich in fibrinolysin
	f. Cremasteric muscle
	g. Masses of erectile tissue

Table III

Column (A)	Column (B)
1. Efferent ductules	a. Opens in prostatic urethra
2. Rete testis	b. Secrete some glycoproteins
3. Vas deferens	c. Found in mediastinum testis
4. Epididymis	d. 20 tubules
5. Ejaculatory duct	e. 65cm in length
	f. Its musculosa is formed of 3 layers
	g. Thin and milky secretion

Table IV

Column (A)	Column (B)
1. Pale type A spermatogonia	a. Haploid no of chromosome
2. Dark type A spermatogonia	b. Largest spermatogenic cell
3. Type B spermatogonia	c. Cycling cells
4. Primary spermatocyte	d. Diffuse endocrine cells
5. Secondary spermatocyte	e. Reserve cells
	f. Mitosis, growth & differentiation
	g. Undergo metamorphosis

Table V

Column (A)	Column (B)
1. Androgen binding protein	a. Epididymis
2. Reinke's crystalloids	b. Cowper's glands
3. Stereocilia	c. Sertoli cells
4. Erectile tissue	d. Prostate gland
5. Corpora amylacea	e. Leydig cells
	f. Penis
	g. Seminal vesicles

Answers

A. MCQ

1	a	7	b	13	c	19	b	25	b
2	c	8	d	14	c	20	b	26	c
3	c	9	a	15	d	21	c	27	b
4	b	10	c	16	b	22	c	28	c
5	a	11	a	17	b	23	b	29	c
6	b	12	a	18	d	24	b	30	d

B. Problem Solving:

1	2	3	4	5	6	7
c	c	b	d	c	c	d

C. Matching

Table I

1	2	3	4	5
c	e	b	f	d

Table II

1	2	3	4	5
b	e	f	d	a

Table III

1	2	3	4	5
d	c	f	b	a

Table IV

1	2	3	4	5
c	e	f	b	a

Table V

1	2	3	4	5
c	e	a	f	d

The Female Genital System

(A) Multiple Choice Questions (MCQs)

Choose the best answer:

- 1. The following is a histological feature of the ovary:**
 - a. Germinal epithelium is formed of simple columnar cells.
 - b. Spindle shaped fibroblasts are observed in the stroma.
 - c. Corpora are found in the medulla.
 - d. Tunica albuginea covers the germinal epithelium.
- 2. Primordial follicles:**
 - a. Are formed after puberty.
 - b. Contain a secondary oocyte with haploid number of chromosomes.
 - c. Are surrounded by a single layer of cuboidal cells.
 - d. Contain a primary oocyte of about 30 micrometer in diameter.
- 3. Primary Unilaminar follicles:**
 - a. Are present at birth.
 - b. Develop at puberty under effect of LH.
 - c. Contain primary oocyte of about 25 micrometer in diameter.
 - d. Are surrounded by single layer of cuboidal cells.
- 4. Zona pellucida:**
 - a. Is formed by both the oocyte & the granulosa cells.
 - b. Is a thick collagen layer.
 - c. Starts to appear in the unilaminar primary follicle.
 - d. It is formed of cuboidal shaped cells.
- 5. The secondary follicle:**
 - a. Is also known as the pre-antral follicle.
 - b. Has a primary oocyte reaching 150 micrometer in diameter.
 - c. Has theca folliculi differentiated into theca interna & externa.
 - d. Contains secondary oocyte with haploid number of chromosomes.
- 6. Theca folliculi cells have the following feature:**
 - a. They are first seen in primary unilaminar follicles.
 - b. The theca externa can be clearly marked from the surrounding stroma.
 - c. They are separated from granulosa cells by a basement membrane.
 - d. Both theca interna and externa cells have a secretory function.

7. **Liquor folliculi is secreted by:**
 - a. Theca externa cells.
 - b. Theca interna cells.
 - c. Zona pellucida.
 - d. Granulosa cells.

8. **Aromatase enzyme is secreted by:**
 - a. Zona pellucida.
 - b. Granulosa cells.
 - c. Theca interna.
 - d. Theca externa.

9. **The type of junction between cytoplasmic processes of corona radiata cells and microvilli from the oocyte is:**
 - a. Tight occluding.
 - b. Desmosome.
 - c. Adherens.
 - d. Gap.

10. **Concerning the corpus luteum, which of the following statements is true?**
 - a. When it is degenerated, it changes into atretic follicle.
 - b. It is concerned with the secretion of relaxin hormone.
 - c. It is formed under effect of LH.
 - d. It is embedded in the ovarian medulla.

11. **The second meiotic division is completed:**
 - a. Just after ovulation.
 - b. Just before ovulation.
 - c. After fertilization .
 - d. At puberty.

12. **Mucosa of the fallopian tube is lined by:**
 - a. Simple columnar partially ciliated epithelium.
 - b. Simple cubical epithelium.
 - c. Simple columnar non ciliated epithelium.
 - d. Pseudostratified columnar ciliated epithelium.

13. **Concerning the endometrium which statement is true?**
 - a. It is lined by simple columnar non ciliated epithelium.
 - b. It contains compound tubular glands.
 - c. Its superficial zone is supplied by straight arteries.
 - d. Its basal zone remains unchanged during the menstrual cycle.

14. **Concerning the myometrium which statement is true?**
 - a. It is hormone insensitive.
 - b. During pregnancy it undergoes hyperplasia & hypertrophy.
 - c. It is formed of three well-defined layers.
 - d. Stratum supravasculare is the thickest layer.

- 15. The proliferative phase of the menstrual cycle:**
- a. At its end the endometrium is about 5 mm thick.
 - b. Shows regenerated straight glands.
 - c. Shows glands with secretory function.
 - d. It starts from day 14 of menstrual cycle till onset of next bleeding.
- 16. The secretory phase of the menstrual cycle:**
- a. Shows straight and short endometrial glands.
 - b. Has an endometrial thickness of 2-3 mm.
 - c. Is under the control of FSH and estrogen hormones.
 - d. The stroma shows decidual reaction.
- 17. Concerning the cervix of the uterus:**
- a. Endocervix is lined by non-keratinized stratified squamous epithelium.
 - b. Ectocervix is lined by mucus secreting simple columnar cell.
 - c. The cervical mucosa is shed during the menstrual cycle.
 - d. Its stroma is formed of dense C.T. with few smooth muscle fibres.
- 18. The cervical mucus:**
- a. In the proliferative phase is less viscous.
 - b. In secretory phase it is abundant.
 - c. In secretory phase under estrogen control it is more viscous.
 - d. Originates from goblet cells in the lining epithelium of the cervix.
- 19. The vagina:**
- a. Under the effect of progesterone, its epithelium synthesizes glycogen.
 - b. Has a lumen with alkaline pH.
 - c. Has lamina propria & adventitia rich in elastic fibres.
 - d. Its lining epithelium secretes mucus.
- 20. Concerning the vagina, which statement is correct?**
- a. It is about 15-20 cm long.
 - b. It has a muscularis mucosa.
 - c. It is lined by stratified columnar epithelium.
 - d. The mucus lubricating the vagina comes from the cervical glands.
- 21. Bartholin's gland:**
- a. Is mucus secreting gland.
 - b. Is situated in the labia majora.
 - c. Is situated on either sides of the clitoris.
 - d. It is a serous secreting gland.

22. The fetal part of the placenta is formed by:
- Decidua capsularis.
 - Decidua parietalis.
 - Decidua basalis.
 - Chorionic villi.
23. Concerning Hofbauer's cells:
- They are found in the trophoblast layer.
 - They are large immotile cells.
 - They are found in intervillous spaces.
 - They are phagocytic cells.
24. Concerning the syncytiotrophoblast, which statement is correct?
- It is a continuous single layer of cuboidal cells.
 - Its cells have pale- stained cytoplasm.
 - It has a secretory function.
 - It disappears by the end of the 4th month of pregnancy.
25. The cytotrophoblast:
- Is formed of pale-stained cuboidal cells.
 - Is responsible for secretion of placental hormones.
 - Is formed of cells with non-clear cell boundaries.
 - Is the outer layer of the trophoblast.
26. Regarding tubal pregnancy the following is true:
- Abdominal pain and shock are rarely encountered if perforation occurred.
 - It is a type of ectopic pregnancy.
 - It is a physiological phenomenon.
 - Pregnancy succeeds.
27. The placental hormone responsible for softening and dilatation of the uterine cervix is:
- Human chorionic gonadotropin.
 - Relaxin.
 - Chorionic thyrotropin.
 - Human chorionic somatomammotropin.
28. The following is true about the vagina:
- Lining cells do not desquamate.
 - The lining cells synthesize mucus.
 - The elastic fibers are sparse in lamina propria.
 - Elasticity is important during labour.

29. **Prolactin hormone is:**
- a. Responsible for enlargement of breasts at puberty.
 - b. Responsible for start of milk secretion from alveoli.
 - c. Responsible for contraction of myoepithelial cells surrounding the breast alveoli.
 - d. Responsible for hypertrophy of myometrial muscles during pregnancy.
30. **The hormone responsible for contraction of myoepithelial cells surrounding the breast alveoli is:**
- a. Prolactin.
 - b. Oxytocin.
 - c. Human chorionic somatomammotropin.
 - d. Relaxin.
31. **The mammary gland is**
- a. Compound alveolar gland.
 - b. Holocrine gland.
 - c. Compound tubuloalveolar gland.
 - d. Merocrine gland.
32. **Concerning Montgomery gland, which statement is correct?**
- a. It is a modified sweat gland.
 - b. It is present in the parenchyma of the mammary gland.
 - c. It increases in size with pregnancy.
 - d. It is an endocrine gland.
33. **Lactiferous sinus in the mammary gland is lined by:**
- a. Simple cuboidal epithelium.
 - b. Simple columnar epithelium.
 - c. Pseudostratified columnar epithelium.
 - d. Stratified cuboidal epithelium.
34. **Concerning mammary gland alveoli, which statement is true?**
- a. They are lined by simple cubical cells.
 - b. They are present in the resting (non-lactating) mammary gland.
 - c. They proliferate from the ends of the ducts.
 - d. By EM, they show few organelles.
35. **The accumulation of adipose tissue and excessive branching of the duct system in mammary gland at puberty is under effect of:**
- a. Progesterone.
 - b. Estrogen.
 - c. Prolactin.
 - d. LH.

36. Breast cancer:

- a. It arises mainly from the alveoli.
- b. It ranks first in Egypt and Worldwide.
- c. Cannot be early identified by self-examination.
- d. Cannot spread to other sites.

(B) Problem solving:

1. A 30-year- old pregnant female at the third trimester, she complained from vaginal bleeding and uterine contractions. She was diagnosed to have Placenta previa and she necessitates cesarean section at time of delivery. This case might result from:
 - a. Implantation of placenta in the upper anterior portion of the uterus.
 - b. Implantation of placenta in the lower portion of the uterus.
 - c. Implantation of placenta in the upper posterior portion of the uterus.
 - d. Implantation of placenta in the abdominal cavity.
2. A 42 -year- old unmarried female complained from heavy and prolonged menstrual periods, low back pain and also had enlargement of uterus. She also suffered from iron deficiency anemia. She was diagnosed to have fibroids. This case might be due to:
 - a. Benign tumor derived from epithelium of endometrium.
 - b. Benign tumor derived from smooth muscles of myometrium.
 - c. Malignant tumor derived from glands of endometrium.
 - d. Benign tumor derived from connective tissue of the perimetrium.
3. A 32-year- old married female complained from, irregular periods, severe acne and extra hair on face and body. She was unable to achieve pregnancy after two years of marriage. Pelvic examination revealed enlarged cystic ovaries & was diagnosed to have Polycystic ovary syndrome. Which statement is true concerning this syndrome:
 - a. It usually affects one ovary only.
 - b. Results in female sex hormones imbalance with arrested primary follicles.
 - c. It is usually accompanied by regular menstrual cycles.
 - d. Could result from excessive production of androgen.
4. A 42-year- old female complained of painless hard mass in her left breast also she experienced breast skin retractions. She also had left enlarged axillary lymph nodes. She was diagnosed as a case of breast cancer. This case might be due to:
 - a. Uncontrolled proliferation of the fat cells in stroma.
 - b. Uncontrolled proliferation of the epidermal cells of the areola.
 - c. Uncontrolled proliferation of the epithelial cells of the alveoli.
 - d. Uncontrolled proliferation of the epithelial cells of the lactiferous ducts

5. A 26- year- old female complaining of menstrual irregularity. Pelvic ultrasound revealed polycystic ovaries. These multiple cysts are:
- Arrested unilaminar primary follicles.
 - Arrested secondary follicles.
 - Arrested primordial follicles.
 - Arrested preantral follicles.
6. A 64 –year-old female complaining of recurrent infection of vagina. She visited the gynecology clinic. She was examined and vaginal acidity test was done. Which result is expected?
- High pH due to increased glycogen synthesis.
 - Low pH due to diminished estrogen level.
 - Low pH due to diminished progesterone level.
 - High pH due to diminished estrogen level.
7. A pregnant female 32 years old has undergone an ultrasonography on the 5th week. It revealed the gestational sac inside the fallopian tube and was diagnosed as tubal ectopic pregnancy. This pregnancy has to be terminated rapidly for fear of perforation because:
- The fallopian tube has a thin musculosa.
 - Tubal mucosa is highly folded.
 - The tubal secretory cells produce insufficient nutritive fluid.
 - The fallopian tube has a narrow lumen.
8. A young female 27 years old has performed a premenstrual endometrial biopsy at day 26 of the menstrual cycle as an investigation for infertility. Which of the following histological findings could indicate that she had hormonal imbalance ending in failure of ovulation:
- Endometrial glands are long with their cells accumulating glycogen in their basal parts.
 - Endometrial glands appear straight with narrow lumen.
 - Spiral arteries are long, coiled and congested.
 - Endometrial glands are highly tortuous, and glycogen accumulated in the apical parts of their cells.

(C) Match Column (A) with Column (B):

Table I

Column (A)	Column (B)
1. Ovarian cortical stroma	a. Glycoprotein layer
2. Antrum	b. No secretory function
3. Theca interna	c. Origin of theca folliculi cells
4. Theca externa	d. Main source of estrogen
5. Zona pellucida	e. Secondary follicles
	f. Germinal epithelium
	g. Tunica albuginea

Table II

Column (A)	Column (B)
1. Hofbauer cells	a. Accumulate glycogen
2. Granulosa-lutein cells	b. Macrophages
3. Vagina	c. disappears by end of 4th month
4. Cytotrophoblast	d. secretes placental hormones
5. Syncytiotrophoblast	e. Secrete progesterone & inhibin
	f. Simple squamous epithelium
	g. Germinal epithelium

Table III

Column (A)	Column (B)
1. Endocervix	a. Simple cuboidal epithelium
2. Ectocervix	b. Transitional epithelium
3. Oviducts	c. Simple columnar partially ciliated
4. Lactiferous Sinus	d. non-kerat. stratified squamous epith.
5. Small lactiferous ducts	e. Simple columnar mucus-secreting cells
	f. Stratified cuboidal epithelium
	g. Germinal epithelium

Table IV

Column (A)	Column (B)
1. Menstrual phase	a. Straight arteries
2. Proliferative phase	b. Straight endometrial glands
3. Secretory phase	c. Spiral arteries
4. Functional zone	d. Decidual reaction
5. Basal zone	e. Destruction of functional zone
	f. Myometrium
	g. Perimetrium

Table V

Column (A)	Column (B)
1. Synthesizes glycogen	a. Bartholin's gland
2. Mucus secretion	b. Granulosa cells
3. Human chorionic gonadotropin	c. Montgomery gland
4. Aromatase enzyme	d. Vaginal epithelium
5. Modified Sebaceous gland	e. Syncytiotrophoblast
	f. Lactogenic activity
	g. High pH

Answers

A. MCQ

1.	b	7.	d	13.	d	17.	d	23.	d	29.	b	35.	b
2.	d	8.	b	14.	b	18.	a	24.	c	30.	b	36.	b
3.	d	9.	d	15.	b	19.	c	25.	a	31.	c		
4.	a	10.	c	16.	d	20.	d	26.	b	32.	c		
5.	c	11.	c			21.	d	27.	b	33.	d		
6.	c	12.	a			22.	d	28.	d	34.	c		

B. Problem Solving:

1	2	3	4	5	6	7	8
b	b	d	d	b	d	a	b

C. Matching

Table I

1	2	3	4	5
c	e	d	b	a

Table II

1	2	3	4	5
b	e	a	c	d

Table III

1	2	3	4	5
e	d	c	f	a

Table IV

1	2	3	4	5
e	b	d	c	a

Table V

1	2	3	4	5
d	a	e	b	c

مها مبارك للطباعة
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